LITERATURE MAP · STATUS 23 SEPTEMBER 2026

Superfluidity of Bose–Einstein condensates in lattices, modulated potentials and two dimensions

What the observables certify

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Abstract. A Bose–Einstein condensate in an optical lattice, a quasi-periodic or speckle potential, or a quasi-two-dimensional trap is superfluid only in part, and the superfluid part is never imaged directly. It is inferred from interference patterns after release, from Bragg and modulation spectra, from the response to a moving obstacle or to a phase twist, and from the decay of coherence with distance. This literature map collects the reference measurements on bosonic gases that define the field: critical velocities from stirring, Bogoliubov spectra from Bragg spectroscopy, the Berezinskii–Kosterlitz–Thouless transition in trapped and uniform two-dimensional gases, the superfluid–Mott-insulator transition, Anderson localisation and the Bose glass, and the recent measurements of the superfluid fraction in modulated gases. It sorts them by what each observable certifies. The map argues that interference peaks and their visibility certify phase coherence, not superfluidity, and that a superfluid fraction needs a response measurement: sound anisotropy, Josephson dynamics or, in a calculation, a phase twist. Leggett's bound is explained as the reason a density modulation suppresses the superfluid fraction. The consequences are drawn for the phase diagram of the twisted-bilayer optical lattice, which rests on peak visibilities. Every number traces to a checked source.

Keywords: Bose–Einstein condensate; superfluid fraction; Landau criterion; Leggett bound; Bogoliubov spectrum; Bragg spectroscopy; two-dimensional Bose gas; Berezinskii–Kosterlitz–Thouless transition; Bose–Hubbard model; Bose glass; time of flight; twisted-bilayer optical lattice

At a glance

10	29	8	192	192	22
chapters	pages of text	figures	sources	sources checked	open questions

Status: complete draft, first edition. Literature status 23 September 2026; checks of sources, numbers and quotations carried out on 23 September 2026 with AI assistance; see "How this map was made" on page i.

The problem

A Bose–Einstein condensate in a periodic, modulated or disordered potential is superfluid only in part, and that part has to be inferred from what can be measured: the interference and momentum patterns after release, the spectrum of excitations, the response to a stirrer or to a phase twist, and the decay of coherence with distance. The question is how the superfluid fraction, the Bogoliubov spectrum and the coherence of a weakly interacting condensate change when the potential removes translational invariance, by a lattice, by quasi-periodic or speckle disorder, or by squeezing the gas into two dimensions, and what each observable certifies about the phase of the gas.

It matters because these experiments, in Paris, Cambridge, Florence, Palaiseau, Munich, NIST and at the Technion, are the reference measurements against which any theory of a modulated Bose gas, including the twisted-bilayer one [1], must be tested. It is hard because superfluidity has several non-equivalent definitions (Landau's critical velocity, the non-classical response to a moving frame, phase stiffness, Leggett's bound), because a two-dimensional gas has no true condensate and its coherence is algebraic (Berezinskii–Kosterlitz–Thouless, BKT), because localised single-particle states in disorder or in narrow bands coexist with delocalised atoms, and because a time-of-flight image reflects the in-trap momentum distribution only in limits [2].

Established: Bogoliubov theory and its Bragg-spectroscopy confirmation [3, 4]; Landau's critical velocity measured by stirring [5, 6]; BKT in trapped and uniform 2D gases [7, 8]; Anderson localisation of expanding condensates in speckle and bichromatic potentials [9, 10]; Leggett's bound saturated in modulated gases and lattices [11, 12]; a Bose glass observed in an optical quasicrystal [13]. Not yet found in the literature checked for this map: a single account of which observable certifies which phase in modulated and disordered quasi-2D condensates, and of how localised excitations and band structure show up in interference and spectroscopy. Progress would be a checked dictionary from observables to phases, anchored in the measured numbers.

How this map was made: scope, method and conventions

Scope. Bosonic condensates in optical lattices, in quasi-periodic and speckle potentials and in quasi-two-dimensional geometry, and the observables that probe them (images, coherence, spectroscopy, stirring, sound, superfluid fraction). Supersolids enter only through one superfluid-fraction method; fermions and electronic systems are out of scope.

Literature cutoff. 23 September 2026. Preprints are included and marked as such; published versions are cited where they were found.

How it was compiled. The starting documents were the project's earlier map on twisted-bilayer optical lattices (161 checked sources, of which 85 are reused here) and its briefing. Three research agents extended them on the Landau criterion and critical velocity, Bogoliubov spectra and Bragg spectroscopy, 2D gases, lattice superfluids and Bose–Hubbard observables, disorder and interference. We looked up every source in a primary record (arXiv, publisher page, group copy) in nine checking rounds with AI assistance. Of the 192 sources the map cites, 55 were confirmed or corrected, 137 partly verified and 0 not verifiable.

Numbers. Each number and quotation traces to a line of the project's facts ledger. Quotations are verbatim from the checked source; numbers confirmed only in part carry "as reported" or "as quoted".

Conventions. $E_{\mathrm{R}} = \hbar^2 k^2 / 2m$ is the recoil energy of the lattice named in each case (sources write E_r or E_{rec}); J nearest-neighbour hopping (t where a source uses it), U on-site interaction; $\tilde{g} = \sqrt{8\pi}\, a / \ell_z$ the two-dimensional coupling, with ℓ_z the oscillator length of the tight axis; $D = n\lambda^2$ the phase-space density; f_{s} the superfluid fraction; c_{s} the sound speed; v_{c} the critical velocity.

In one page: the field and the argument

Cold-atom experiments have measured most of what the textbook says a superfluid does, in gases whose density can be imaged and whose potential is written with light. What they rarely measure is the superfluid fraction itself, and many phase diagrams of lattice and disordered gases rest on interference patterns instead. We argue in five steps that this matters.

1 **The reference measurements are in place.** Stirring gives critical velocities well below the sound speed: $v_c \approx 1.6\,\mathrm{mm/s}$ against $c_s = 6.2\,\mathrm{mm/s}$ at the peak density of a three-dimensional condensate [5], (0.07 ± 0.02) to $(0.11 \pm 0.02)\,c_s$ depending on density [14], and $v_c/c_s = 0.3$–0.6 in two dimensions [6]. The BKT constants are known, $C = 380 \pm 3$ in $D_c = \ln(C/\tilde{g})$ [15, 16] and the jump $n_s\lambda^2 = 4$ that sound measurements confirm [8]. The superfluid–Mott-insulator point sits at $(U/J)_c = 29.34(2)$ in three dimensions by quantum Monte Carlo [17], against a measured 34.2(2.0) [18], and at $(J/U)_c = 0.05974(3)$ in two [19].

2 **Interference measures coherence, not superfluidity.** In quantum Monte Carlo "sharp peaks are not a reliable diagnostic of superfluidity": they survive above T_c, where superfluid density and condensate fraction vanish [20]. A quasi-2D gas can be bimodal and "quasi-condensed but is not superfluid" [21]. For coherent ^{87}Rb clouds the far field needs about 100 ms of flight, against the 20 ms typically used [2]. Visibility and peak width do change at the Mott transition, which is why they locate critical points [22, 23]; they do not say how much of the gas flows.

3 **Superfluidity needs a response.** Sound along and across a one-dimensional modulation gives $f_{s,x} = c_x^2/c_y^2$, and for a weakly interacting condensate Leggett's bound becomes an identity [11]. In a lattice "the speed of sound saturates Leggett's bound" [12]. In a triangular lattice a static and a dynamic method agree and give $f_s^{(yy)} = 0.82(2)$ [24]; Josephson oscillations measure f_s in a supersolid [25]. In a calculation the response is a phase twist or a winding number, which finds $f_s \approx 0$ in a quasicrystal Bose glass while the compressibility changes smoothly [26].

4 **Leggett's bound explains the suppression.** The bound uses the density profile alone. For a modulation along x it reads $f_s \le [\bar{n}\langle 1/n\rangle_x]^{-1}$, the one-dimensional case of the brackets of Pérez-Cruz *et al.* [27, 28]: flow along x must cross every low-density region, so a density that nearly vanishes somewhere forces f_s down, whatever the band structure.

5 **The twisted-bilayer diagram needs a response measurement.** The one twisted-bilayer experiment labels its phases by time-of-flight peaks at the primary and the moiré momenta; SF-II keeps only the primary ones and is called "superfluid only with short-range coherence" [1]. By steps 2 and 3, these labels certify coherence at two length scales. How much of the gas flows is not known: no superfluid-fraction or sound measurement on a moiré gas was found in the literature checked for this map. The in-situ images already resolve the 4.35 μm moiré density [1], so Leggett's upper bracket could be evaluated from existing data.

What a calculation should report

Alongside the momentum distribution, a theory of a modulated condensate should give the quantities a response measurement returns: c_x and c_y, the f_s tensor from a phase twist, and Leggett's brackets from its own density [11, 24, 28]. Chapter 3 explains how each is measured, and chapter 6 argues the case.

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Terms from superfluidity, two-dimensional gases, optical lattices, disorder and cold-atom measurement, as this map uses them.

How to read this map

Twenty minutes. The problem, the one-page summary, then the key-points boxes at the end of each chapter.

Coming from theory, new to the lab. Chapter 3, the experimental toolbox and its dictionary.

Working on the twisted-bilayer problem. Chapters 6 and 8.

Looking for a research question. Chapter 9.

Boxes: key points, what a result means for the reader, open questions.

1 The topic in brief

What "superfluid" means in a cold-atom laboratory, the three ways a potential removes translational invariance, the experiments that define the field, and where the question of observables comes from.

A Bose–Einstein condensate (BEC) of a dilute gas is the simplest superfluid: "At zero temperature, a Galilean-invariant Bose fluid is expected to be fully superfluid" [11]. In the laboratory superfluidity is never seen directly. It is inferred from a response: a stirrer or a moving lattice that dissipates only above a critical velocity, a slow motion of the potential that part of the gas does not follow, the speed of sound along and across a modulation, or the phase coherence that shows up as interference after the gas is released. The first three probe the definitions of chapter 2; the last probes coherence, and the distinction between the two runs through the whole map.

1.1 Three ways of removing translational invariance

A periodic optical lattice, made by standing waves of far-detuned light, turns the condensate into a Bloch state with a heavier band mass; deep lattices map onto the Bose–Hubbard model and its Mott insulator [29–31]. Disorder, from laser speckle, from two incommensurate lattices or from an optical quasicrystal, localises single-particle states and, with interactions, gives the Bose glass [9, 10, 13, 30, 32]. Squeezing the gas into a plane with $\hbar\omega_z \gg k_B T$, gn removes invariance along z and lets phase fluctuations destroy long-range order in the plane; the transition becomes the Berezinskii–Kosterlitz–Thouless (BKT) transition [16, 33–35]. The three are combined in practice: square lattices on two-dimensional gases [36], a triangular lattice on a uniform plane [24], an eight-fold quasicrystal on a quasi-two-dimensional gas [13], and two twisted square lattices on a pancake condensate [1]. Figure 1.1 crosses the potential types with the observables, and Table 1.1 lists the platforms.

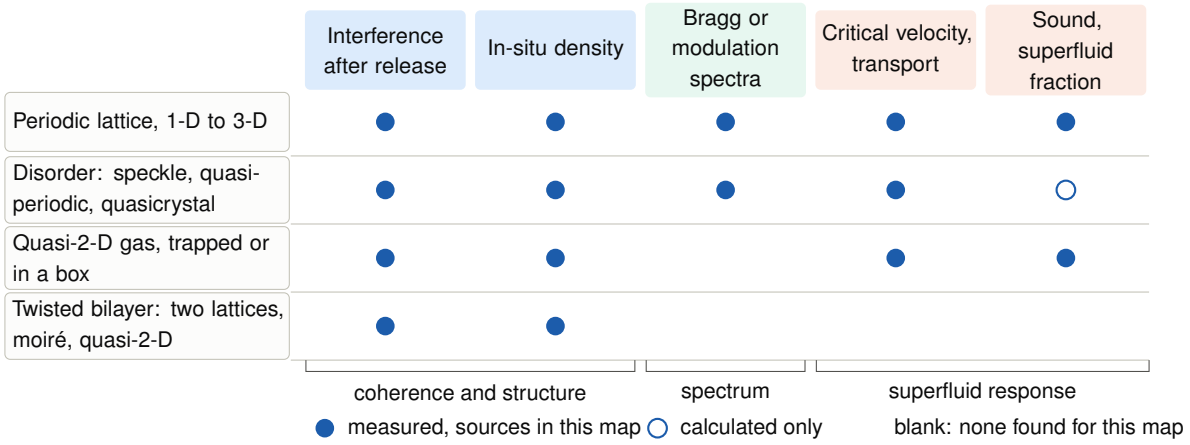


Figure 1.1: Potential types against observables. A filled dot means that at least one experiment among the sources of this map measured the quantity for that potential (Table 1.1 and chapter 5); an open dot means only calculations were found (the superfluid fraction of a Bose glass, from quantum Monte Carlo winding numbers [26]); a blank means none was found in the literature checked for this map, which is not proof of absence. The twisted-bilayer experiment also used microwave spectroscopy of the single-particle bands [1], which is not an excitation spectrum of the gas and is not marked.

1.2 The experiments that define the field

The critical velocity came first. A blue-detuned beam moved through a three-dimensional sodium condensate caused "Strong heating … only above a critical velocity", well below

Table 1.1: Platforms for bosons in modulated potentials. "Dim." is the dimension of the gas or of the modulation as the sources state it. Keys are a selection; chapter 5 gives the rest.

Platform	Potential	Dim.	What is measured	Key sources
Trapped BEC, stirred or Bragg-probed	harmonic trap; moving blue-detuned beam; two Bragg beams	3-D	critical velocity for heating or drag; Bogoliubov spectrum	[3–5, 14]
Trapped quasi-2-D gas	light sheet or planes of a 1-D lattice	2-D	interference of planes; equation of state; critical velocity; coherence exponent	[6, 7, 37, 38]
Uniform 2-D gas	optical box from a mask; single plane	2-D	sound; superfluid-density jump; defects after a quench	[8, 39, 40]
BEC in a 1-D lattice	shallow standing wave along one axis	1-D lattice	effective mass; Josephson oscillations; dissipation and dynamical instability	[41–44]
Deep lattices, SF–MI	cubic or square lattice; arrays of tubes	3-D, 2-D, 1-D	interference visibility; Mott gap; in-situ shells; amplitude mode	[31, 36, 45–47]
Modulated gas, f_s	1-D modulation, 1-D lattice or triangular lattice	2-D gas; 1-D lattice	superfluid fraction from sound, Bragg phonons or the density profile	[11, 12, 24]
Disordered gas	laser speckle; bichromatic lattice	1-D, 3-D	localised expansion; momentum width; modulation spectra; transport	[9, 10, 48–50]
Optical quasicrystal	four 1-D lattices at 45°, eight-fold	2-D	coherence of the diffraction pattern; localisation depth; Bose glass	[13, 51–53]
Twisted-bilayer optical lattice	two square lattices rotated by 5.21°, one per spin state; microwave coupling	quasi-2-D	in-situ moiré density; visibilities of primary and moiré peaks; band spectroscopy	[1]

the sound speed [5, 14]; a persistent current in a ring stopped abruptly when the flow at a barrier exceeded a critical value [54], and the same test was made in a two-dimensional gas [6]. Bragg spectroscopy measured the Bogoliubov spectrum, its phonon part and the Bogoliubov transformation itself [3, 4, 55–57].

In two dimensions the Paris group observed the BKT crossover in a trapped ^{87}Rb gas: "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. Scale invariance and the equation of state followed in Chicago and Paris [37, 58], the critical point was measured against classical-field theory [59], and uniform gases in optical boxes [39] gave sound and, from first and second sound, the superfluid density with its universal jump [8, 40].

In lattices, a condensate in a one-dimensional standing wave showed a heavier effective mass, Josephson oscillations between wells and a velocity above which its motion dissipates [41–43]. Greiner *et al.* deepened a three-dimensional lattice and saw "a transition … from a superfluid to a Mott insulator phase" [31]. Modulation and Bragg spectroscopy then measured the Mott gap, the Bogoliubov bands and the amplitude mode of lattice superfluids [45, 47, 60–62], and quantum-gas microscopes imaged the Mott shells site by site [46, 63].

Disorder entered with Anderson localisation of expanding condensates in speckle and in bichromatic lattices [9, 10], then interacting and Bose-glass regimes [48, 49, 64, 65]. In an eight-fold optical quasicrystal the Cambridge group observed "a Bose-glass-to-superfluid transition" by probing coherence [13, 51, 52], and a preprint follows quenches across that transition [53].

The superfluid fraction itself has been measured since 2023: from the anisotropy of sound

in a two-dimensional gas with a one-dimensional modulation [11], from Bragg phonons in a one-dimensional lattice [12], as a full tensor in a triangular lattice [24], and from Josephson oscillations in a dipolar supersolid [25]. The twisted-bilayer optical lattice closes Table 1.1: two square lattices, one per spin state of ^{87}Rb, rotated by 5.21° and coupled by a microwave, on a quasi-two-dimensional condensate [1].

1.3 Where the question of observables comes from

Most phase diagrams of lattice and disordered Bose gases are read from interference after release. In the quasicrystal, "A high-contrast interference pattern composed of sharp peaks indicates the presence of long-range phase coherence, the signature of the superfluid phase" [13], and the twisted-bilayer phases are labelled by which families of peaks are present [1]. Three results show that this reading needs care. Quantum Monte Carlo (QMC) calculations for the three-dimensional Bose–Hubbard model found that "sharp peaks are not a reliable diagnostic of superfluidity", because they survive above the critical temperature where the superfluid density and the condensate fraction vanish [20]. A quasi-two-dimensional gas can be "quasi-condensed but … not superfluid" [21]. And the far field that turns an image into a momentum distribution is reached only after about 100 ms of flight, against the usual 20 ms [2]. Interference measures coherence over some length; superfluidity is a response. Which observable certifies which phase is the question of this map: chapter 3 describes the measurements, chapter 4 the benchmarks, and chapter 6 states the position.

Key points

- Superfluidity is inferred from a response: critical velocity, the part of the gas that does not follow a moving potential, or sound; interference after release measures coherence.
- Translational invariance is removed by a lattice, by disorder or quasi-periodicity, or by squeezing the gas into two dimensions; experiments combine them.
- Direct measurements of the superfluid fraction of modulated condensates exist only since 2023, from two groups using sound and one using Josephson oscillations.
- Sharp interference peaks survive above the critical temperature in QMC, so phase diagrams read from peak visibilities need a response measurement to be read as superfluid.

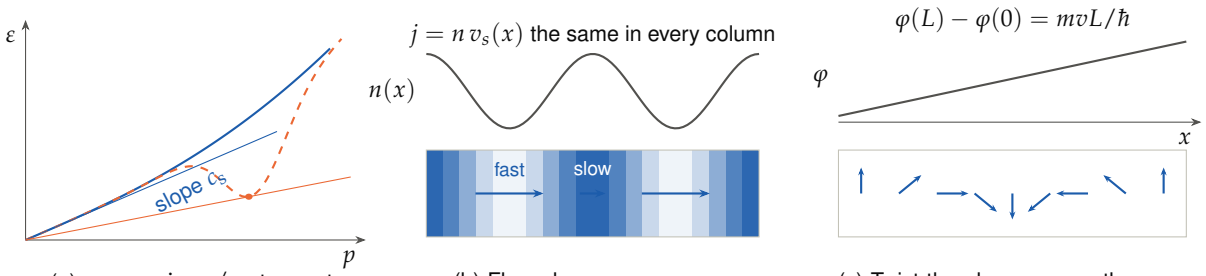
2 Foundations

The definitions behind the observables: Landau's critical velocity, the superfluid fraction and Leggett's bound, the Bogoliubov spectrum, the Bose–Hubbard model and the two-dimensional gas.

We write E_R for the recoil energy of the lattice at hand (experiments write E_r), and J for the hopping (t where a source writes t). Figure 2.1 draws the three pictures of this chapter.

2.1 Landau's criterion and the critical velocity

A fluid flowing at velocity v past an obstacle can lower its energy by creating an excitation of momentum p and energy $\varepsilon(p)$ only if $\varepsilon(p) - pv < 0$. Flow is therefore dissipationless below $v_\mathrm{c} = \min_p \varepsilon(p)/p$, the slope of the tangent from the origin to the spectrum (Fig. 2.1a) [66, 67], a necessary condition for a metastable superflow [68]. For the Bogoliubov spectrum the minimum sits at $p \to 0$ and $v_\mathrm{c} = c_\mathrm{s}$, the sound speed; Steinhauer *et al.* note that the measured linear regime "provides an upper limit for the superfluid critical velocity, by the Landau criterion" [4].



(a) $v_c = \min_p \varepsilon/p$: tangent from the origin; $v_c = c_s$ for Bogoliubov (solid), $v_c < c_s$ with a minimum (dashed)

(b) Flow along x crosses every column; thin columns force a fast flow and cost energy: $f_s \leq [\langle n\rangle\langle 1/n\rangle]^{-1}$

(c) Twist the phase across the sample: $E(v) = E(0) + f_s N m v^2/2$

Figure 2.1: Schematics. (a) Landau's criterion: v_c is the slope of the tangent from the origin to the spectrum. (b) Leggett's column picture (darker is denser): one current through every column, so the flow is fast where the density is low. (c) A phase winding by $mvL/\hbar$ over a length L imposes a flow v; the energy cost defines f_s.

Stirred gases stay well below that limit (Fig. 2.2a). In sodium the crossover came "at a velocity of about 1.6 mm/s", against a sound speed of 6.2 mm/s at the peak density [5]; the onset of a drag force came at $v_c = (0.11 \pm 0.02)\, c_s$ and $(0.07 \pm 0.02)\, c_s$ at the higher and the lower density [14]. In a two-dimensional ^{87}Rb gas $v_c/c_s = 0.3$–0.6, and frictionless flow set in at a local $\mu/k_BT \approx 0.24$, above the predicted 0.15 [6]. A ring's critical velocity is "consistent with dissipation due to the creation of vortex-antivortex pairs" [54]: the obstacle nucleates vortices, and the local sound speed falls where the density does.

In a lattice the reading is not settled. Burger *et al.* measured $m^*/m = 1.18 \pm 0.02$ and a full loss of superfluidity at $v_{\max} = (5.3 \pm 0.5)$ mm/s, against a computed $c_{s,\max} \simeq 5.2$ mm/s, and invoked "the spontaneous emission of elementary excitations" without naming Landau [41]. In a moving lattice Fallani *et al.* found "a discontinuity around $q = 0.55 q_B$" in the lifetime, "unambiguously attributed to the onset of dynamical instability, after a comparison with the predictions of the Gross-Pitaevskii theory" [43]. In three dimensions the critical momentum falls to zero at the Mott transition [18]. Energetic (Landau) and dynamical instabilities have different thresholds in a lattice, and only a model tells which one an experiment crossed.

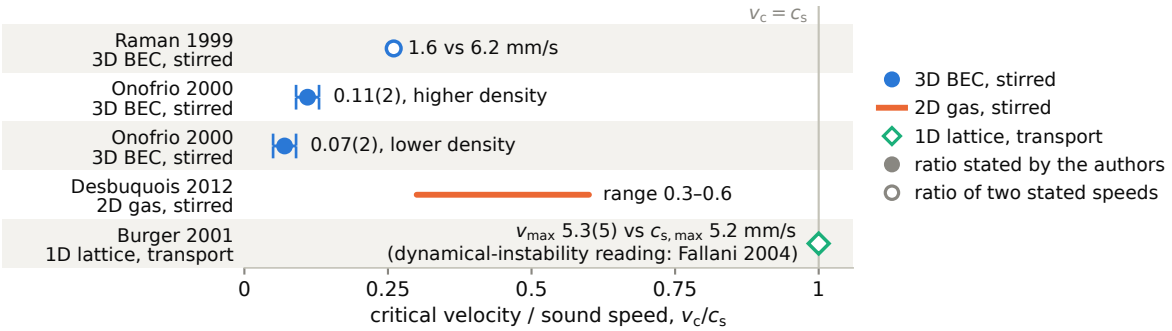
2.2 The superfluid fraction

"The superfluid fraction f of a quantum fluid is defined in terms of the response of the system to a weak and constant drag" [28]. Move the potential slowly at velocity $\mathbf{v}$: the normal part is dragged along, the superfluid part is not, and for N atoms of mass m [28, 69]

$$f_{\alpha\beta} = 1 - \lim_{v\to 0} \frac{\langle \hat{P}_\alpha \rangle}{N m v_\beta}, \qquad E(v) = E(0) + f_s \frac{N m v^2}{2} \quad (v \to 0). \tag{1}$$

The effect is called non-classical translational inertia, and in an anisotropic potential f_s is a rank-2 tensor [70]. The second form, as written by Pérez-Cruz *et al.*, is the one used in calculations: a phase winding by $mvL/\hbar$ across the sample (Fig. 2.1c) costs an energy set by f_s; lattice models call it the helicity modulus [71], and QMC samples it through winding numbers [72]. Sound gives a measurable form. With c_x along and c_y across a one-dimensional modulation, Chauveau *et al.* write $c_x^2 = f_{s,x}/(m\kappa)$, κ the compressibility, hence $f_{s,x} = c_x^2/c_y^2$, and note that for their weakly interacting condensate $f_{s,x}$ coincides exactly with m/m_x^*, the ratio of the bare to the band mass [11]. Taylor and Zaremba's sound speed, $s^2 = (\bar{n}/m^*)\,\partial\mu/\partial\bar{n}$ as it is usually quoted, carries the same band mass [73].

a Measured critical velocities



b Superfluid fraction: what was measured

experiment	method	potential	result as stated
Chauveau 2023	sound anisotropy c_x^2/c_y^2; Leggett formula on the density	2D ^{87}Rb, 1D modulation, $d = 3.93(4)$ µm, $V_0 = 0$–80 nK	Leggett bound exact for 1D modulation (confirmed); f_s plotted, no numeric value given
Tao 2023	Bragg-scattered phonons (sound speed)	1D lattice, $a = 266$ nm, depth up to 10 E_R	"the speed of sound saturates Leggett's bound"; no numeric value given
Christodoulou 2021	first and second sound	2D gas across the BKT transition	superfluid density consistent with BKT, including the universal jump at T_c
Biagioni 2024	Josephson oscillations of single supersolid cells	dipolar supersolid (self-modulated density)	"a relatively large sub-unity superfluid fraction"; no numeric value given
Rabec 2026	density + continuity eq. (static); compressibility + sound (dynamic)	2D ^{87}Rb, triangular lattice, $d = 6.0(1)$ µm, V_0 up to 4.7 μ_0	$f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\, g\rho_0$; both methods agree with GPE and with the Leggett bounds
Gautier 2021 (QMC)	path-integral QMC (winding numbers)	8-fold quasicrystalline lattice	Bose glass: $f_s \approx 0$; $V_c/E_R \simeq 1.76 \pm 0.01$ (weak interactions); smooth compressibility crossover

Values as stated by the cited authors; no recomputation.

Figure 2.2: (a) Measured critical velocities as a fraction of the sound speed. The ratios depend on the obstacle and on the density at which c_s is taken; the lattice point compares a maximal velocity with a computed sound speed, and its authors do not call it a Landau threshold [41]. (b) Measurements of the superfluid fraction: method, potential and result as stated by the authors; the last row is a QMC calculation, not an experiment [26]. No values were recomputed for this map.

2.3 Leggett's bound, explained

Take a density $n(x)$ modulated along x and impose a slow flow along x with a phase that depends on x only, keeping the density fixed. In a steady state the current $j = n(x)\, v_s(x)$ is the same through every slice of constant x (Fig. 2.1b), so $v_s = j/n$: the flow is fast where the gas is thin. The twist fixes the mean velocity, $\langle v_s \rangle = j\langle 1/n \rangle = v$, and the kinetic energy per unit length is $\frac{m}{2}\langle n v_s^2 \rangle = \frac{m}{2} v^2 / \langle 1/n \rangle$. Equating it to $f_s \langle n \rangle m v^2/2$ of Eq. (1) gives $1/(\langle n \rangle \langle 1/n \rangle)$. The true twisted ground state lies at or below this phase-only trial state in energy, so the value is an upper bound [27, 28]:

$$f_s \leq \frac{1}{\langle n \rangle \, \langle 1/n \rangle}. \tag{2}$$

Chauveau *et al.* state that "the bound reduces to an identity in the case of a weakly interacting BEC subject to a 1D periodic potential" [11]: a condensate is a single orbital whose only freedom is the phase. Because $\langle n \rangle \langle 1/n \rangle \geq 1$, with equality only for a uniform density, any modulation lowers f_s, and the harmonic mean $1/\langle 1/n \rangle$ is dominated by the thinnest columns: a narrow region where n nearly vanishes chokes the flow. This is why a density modulation suppresses the superfluid fraction of a condensate that is fully superfluid when uniform.

In two dimensions a density $n(x,y)$ gives two brackets for a twist along x [28]:

$$f^{+} = \frac{1}{\bar{n}} \frac{1}{\left\langle 1/\langle n\rangle_y \right\rangle_x}, \qquad f^{-} = \frac{1}{\bar{n}} \left\langle \frac{1}{\langle 1/n\rangle_x} \right\rangle_y, \qquad f^{-} \le f_{\mathrm{s}} \le f^{+}. \tag{3}$$

One way to read them: f^{+} comes from a trial phase that depends on x only, one velocity across a whole column, so only the column average $\langle n\rangle_y$ enters Eq. (2); f^{-} lets each line of constant y carry its own current, as channels in parallel, and is the dual bound built from trial currents. Both follow from the density alone. "The bracketing is tighter for a 2D optical potential than for a grid of columns" [28]; the triangular-lattice measurement is "in good agreement … with the Leggett bounds" [24]. Two preprints extend the bound to the direction of the twist [74] and to strongly interacting bosons [75].

2.4 Bogoliubov spectrum and sound

Linearising the Gross–Pitaevskii equation [76, 77] about a uniform condensate of density n gives $\varepsilon(p) = [c_{\mathrm{s}}^2 p^2 + (p^2/2m)^2]^{1/2}$ with $mc_{\mathrm{s}}^2 = gn$: phonons below the inverse healing length, free particles above, and a quantum depletion from the same theory [67, 78]. Bragg spectroscopy confirmed it. The mean-field shift was $(0.54 \pm 0.07)\, n_0 U/h$, "in excellent agreement with the prediction of $4n_0U/7h$" [3]; the phonon regime showed a suppressed structure factor [55]; the full spectrum gave $c_{\mathrm{eff}} = 2.0 \pm 0.1\,\mathrm{mm/s}$ with $\mu/h = 2.02 \pm 0.09\,\mathrm{kHz}$ (arXiv version) [4]; and the quasiparticle was seen as a superposition of $\pm p$ [56, 57]. In a lattice the spectrum splits into Bogoliubov bands with their own dynamic structure factor [79], and Krämer *et al.* express the sound velocity through the effective mass and the compressibility of the lattice condensate [80]; chapter 4 lists where the measured lattice spectra depart from Bogoliubov.

2.5 The Bose–Hubbard model

For a deep lattice the atoms occupy the lowest Wannier orbital w of each site, and [29, 30]

$$H = -J \sum_{\langle ij\rangle} b_i^{\dagger} b_j + \frac{U}{2} \sum_i n_i(n_i - 1) - \mu \sum_i n_i, \tag{4}$$

with $n_i = b_i^{\dagger} b_i$ and $\langle ij\rangle$ nearest neighbours; both parameters are set by the laser light [29]. J is the tunnelling through the barrier and falls exponentially with depth, $J \approx (4/\sqrt{\pi})\, E_{\mathrm{R}}\, s^{3/4} e^{-2\sqrt{s}}$ with $s = V_0/E_{\mathrm{R}}$ [81]; $U \propto g \int |w|^4$ is the interaction of two atoms in one orbital and grows slowly as the orbital shrinks. Deepening the lattice raises U/J by orders of magnitude, from a superfluid (SF) with long-range phase coherence to a Mott insulator (MI) with "exact numbers of atoms … localized at individual lattice sites, with no phase coherence across the lattice" and "a gap in the excitation spectrum" [31].

At unit filling mean-field theory gives $(U/J)_c = 5.8z$ for z neighbours [81], 34.8 on the cubic lattice [18]. QMC gives $(U/t)_c = 29.34(2)$ in three dimensions [17] and $(J/U)_c = 0.05974(3)$, that is $(U/J)_c \approx 16.74$, on the square lattice [19]; the density-matrix renormalisation group (DMRG) gives $t_c = 0.277 \pm 0.01$ in units of U in one dimension, $(U/t)_c \approx 3.61$ [82]; mean field misses more as fluctuations grow at lower dimension [83–85].

Four observables tell SF from MI. Interference peaks and their visibility: "at a potential depth of around 13 E_r the interference maxima no longer increase in strength … at a potential depth of 22 E_r no interference pattern is visible at all" [31], and a residual visibility survives in the Mott phase [22, 86]. The gap, a discrete spectrum in lattice-modulation spectroscopy with its first resonance near U [45]. Integer filling, imaged in situ as plateaus and shells site by site [46, 63]. Number fluctuations, suppressed in the insulator with the local compressibility [87],

and noise correlations of the expanding cloud [88]. The model has limits: continuous-space QMC finds that at 15 E_R "the Bose–Hubbard approximation fails to describe but the first Mott lobe" of hexagonal potentials, with density-assisted tunnelling beyond it [89]; such non-standard terms are reviewed in [90], and measured number-dependent shifts reveal effective multi-body interactions [91].

2.6 Quasi-two-dimensional gases and the BKT transition

A gas is quasi-two-dimensional when "the frequency of the tight confinement ω_0" is "much larger than the gas temperature T and the mean-field interparticle interaction $n_0 g$" [33]. Interactions then enter through $\tilde{g} = \sqrt{8\pi}\, a_s/a_z$, with $a_z = \sqrt{\hbar/m\omega_z}$, which "varies between $\sim 10^{-1}$ and $\sim 10^{-2}$" in current experiments [16]: $\tilde{g} \approx 0.1$ in the Paris equation of state [37], 0.16(1) in the uniform box of Ville *et al.* [40], 0.158(4) in the triangular lattice [24]. The sound speed is $c_s = \hbar\sqrt{\tilde{g}n}/m$, about 1.6 mm/s at $n = 50\,\mu\mathrm{m}^{-2}$ [6].

In an infinite plane at $T > 0$ phase fluctuations destroy long-range order and the first-order coherence decays algebraically [34]; the superfluid disappears when vortex pairs unbind [35]. For the weakly interacting gas the critical phase-space density is $D_c = (n\lambda^2)_c = \ln(380/\tilde{g})$, with 380 ± 3 from classical-field Monte Carlo [15, 16, 92]; the critical chemical potential carries a second constant, 13.2 ± 0.4 [92]. The superfluid density jumps by the universal amount $n_s\lambda^2 = 4$ [16, 93], and first and second sound in a uniform gas agree with BKT theory "including the prediction of a universal jump in the superfluid density" [8]. The trapped critical point matches classical-field predictions "without any free parameters" [59], while an interferometric coherence exponent $\eta_c = 0.17(3)$ lies below the universal 0.25 [38]. In a finite trapped gas a true condensate reappears well below T_c, with a quasicondensate in between [33], and a bimodal, quasi-condensed gas need not be superfluid [21].

Key points

- Landau's criterion bounds v_c from above; stirred gases give v_c/c_s from below 0.1 to 0.6, and lattice experiments are read either as Landau or as dynamical instability.
- f_s is the part of the gas that does not follow a slowly moving potential: an energy cost under a phase twist, and $f_{s,x} = c_x^2/c_y^2$ from sound.
- Leggett's bound $[\langle n\rangle\langle 1/n\rangle]^{-1}$ is an identity for a weakly interacting condensate in a one-dimensional modulation; in two dimensions the density gives two brackets.
- Bose–Hubbard: $(U/J)_c \approx 29.3$, 16.7 and 3.6 in three, two and one dimensions, below mean field; BKT: $D_c = \ln(380/\tilde{g})$ and $n_s\lambda^2 = 4$.

3 Context: the experimental toolbox, for theorists

How a quasi-two-dimensional or lattice condensate is prepared, probed and imaged, and which theoretical quantity each measured signal stands for.

Every experiment here traps a gas, shapes its potential, probes it and takes an image (Fig. 3.1); Table 3.1 pairs each quantity with its signal and its pitfall.

3.1 Making a quasi-two-dimensional gas

The gas is quasi-two-dimensional when the tight confinement freezes the motion along z. Petrov *et al.* require "the frequency of the tight confinement ω_0 to be much larger than the gas temperature T and the mean-field interparticle interaction $n_0 g$" [33]; Hadzibabic and Dalibard add $a_z \gg \xi, \lambda/\sqrt{2\pi}$ for the oscillator length a_z, the healing length ξ and the thermal wavelength λ [16]. Collisions stay three-dimensional, and the scattering length a enters through $\tilde{g} = \sqrt{8\pi}\, a/\ell_z$, which "varies between $\sim 10^{-1}$ and $\sim 10^{-2}$" in experiments [16]: $\tilde{g} \approx 0.1$ at $\omega_z/2\pi = 1.9(2)$ kHz in the Paris equation-of-state work [37], $\tilde{g} = 0.158(4)$ at $\omega_z/2\pi = 4.1(2)$ kHz in the 2026 triangular-lattice experiment [24].

The tight confinement is made with light. A standing wave along z made of two running beams gives a stack of planes [16], and interfering two planes after release revealed the BKT crossover by matter-wave heterodyning [7]; a vertical double well with about 1 kHz per plane serves the same purpose [38]. An accordion lattice, with spacing tuned "between 11 and 2 μm", loads a single plane and compresses it to $\omega_z/2\pi > 10$ kHz [94, 95]. A beam with an intensity node at $z = 0$ gives $\nu_z = 350$–1500 Hz, and a mask on a second beam cuts a box of 200–900 μm^2 in the plane [39]; a digital micromirror device (DMD) imaged onto the atoms makes such potentials programmable [96]. A box removes the averaging over densities that a harmonic trap imposes, and the Paris sound and superfluid-fraction experiments use one [24, 40]. A review [16] and lecture notes (a preprint) [97] cover this physics.

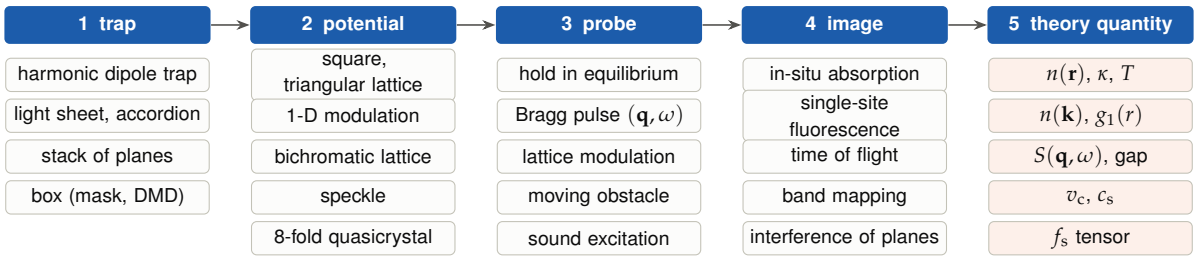


Figure 3.1: The experimental sequence as a theorist meets it. Each column lists the options used by the experiments of this chapter; the last is what the analysis returns. The choice of probe and image fixes which quantity is measured (Table 3.1).

3.2 Imaging: in situ and after time of flight

An in-situ absorption image gives the column density, the density integrated along the line of sight. With the local density approximation (LDA), which treats each point of the trap as a uniform gas at the local chemical potential, one image yields an equation of state; in two dimensions the dimensionless quantities depend on $\mu/k_B T$ only [37]. In a lattice the same route gives a local compressibility, strongly suppressed in the Mott domains, and density fluctuations that act as a thermometer through the fluctuation–dissipation theorem [87]; in-situ density has also followed slow mass transport [98] and quantum-critical scaling with $z = 2.0(3)$, $\nu = 0.54(5)$ [99]. Quantum-gas microscopes resolve single sites. In the Harvard apparatus (lattice period 680 nm) atoms of the $n = 1$ shell are detected with probability 94.9 ± 0.7 % at 16 E_R, and shells up to $n = 4$ appear through parity projection: the microscope sees the occupation modulo two [46]. The Munich microscope takes temperature and entropy from single images and shows "Mott-insulating plateaus with near-zero entropy" [63].

Most lattice experiments image after time of flight (TOF): the potential is switched off and the cloud expands for tens of milliseconds. Only in the far field does the image reproduce the in-trap momentum distribution. Gerbier *et al.* give the far-field time $t_{\mathrm{FF}} = m R_0 l_c/\hbar$, with R_0 the cloud size and l_c the coherence length; for coherent ^{87}Rb samples $t_{\mathrm{FF}} \approx 100$ ms against a typical 20 ms,

so "the far-field condition is usually not met" [2]. In two dimensions a focusing technique gives momentum-space images directly [100]. Ramping the lattice down in 2 ms maps the first band onto the first Brillouin zone [101], and correlations of the density noise in the expanding cloud reveal the order of a Mott insulator [88]. Coherence itself is read from the interference of two condensates [102, 103], from the correlation function, which decays as a Gaussian above T_c and to a plateau below [104], from the visibility and width of lattice TOF peaks [22, 23, 86], and in low dimensions from the statistics of fringe contrast [105].

3.3 Spectroscopy and stirring

Bragg spectroscopy uses two beams with momentum difference $\hbar\mathbf{q}$ and frequency difference ω; the number of atoms transferred measures the dynamic structure factor $S(\mathbf{q},\omega)$. Stenger *et al.* found a mean-field shift of slope $(0.54 \pm 0.07)\, n_0 U/h$ against the LDA prediction $4n_0U/7h$ [3], and at small q the phonon regime suppresses the response [55]. Steinhauer *et al.* measured the whole spectrum, linear then parabolic, with $c_{\mathrm{eff}} = 2.0 \pm 0.1$ mm/s: "The linear regime provides an upper limit for the superfluid critical velocity, by the Landau criterion" [4]; the Bogoliubov transformation itself was observed [56], and a review covers these measurements [57]. In lattices, momentum-resolved Bragg spectra show the band structure and the limits of Bogoliubov theory [44, 61, 79]; the resonance falls faster with depth than Bogoliubov theory predicts [62], and in one-dimensional gases the response "drastically drops" in the insulating regime [60].

Lattice-modulation spectroscopy shakes the lattice depth and measures the energy absorbed [106]. The superfluid shows a broad continuum, the Mott insulator a discrete spectrum with a first resonance near U and a second at (1.91 ± 0.04) times the first [45]; precision versions resolve number-dependent shifts from effective multi-body interactions [91]. In two dimensions the onset of absorption softens towards the critical point, $j_c \simeq 0.06$ from quantum Monte Carlo (QMC): the amplitude ("Higgs") mode [47], which Bragg scattering also detects [107].

Stirring tests Landau's criterion, a necessary condition for the metastability of a superflow [68]. A blue-detuned beam moved through a sodium condensate heated it strongly only above about 1.6 mm/s [5]; in a two-dimensional ^{87}Rb gas a beam of waist 2.0(5) μm gave critical velocities of 0.5–1.0 mm/s [6]. In a ring, flow past a barrier stops at a velocity consistent with the creation of vortex–antivortex pairs [54]. Moving lattices probe the same physics in a band [18, 41, 43].

3.4 Disorder and quasi-periodic potentials

Speckle, laser light passed through a diffuser, is a disorder with a finite correlation length [32, 108]. In one dimension it localises expanding atoms exponentially only when their de Broglie wavelength exceeds "an effective mobility edge" set by the speckle cutoff [9]; in three dimensions localised and diffusive components separate [50] and the mobility edge has been measured [109]. A second, incommensurate lattice gives quasi-periodic disorder (830 and 1076 nm in Florence, as reported [48]), and the critical disorder for localising a non-interacting condensate scales with J [10]. Four one-dimensional lattices at $45(1)^\circ$ with $\lambda = 726$ nm make an eightfold quasicrystal [51], where the two-dimensional Bose glass was observed [13, 52].

3.5 Measuring the superfluid fraction

Chauveau *et al.* imprint a one-dimensional modulation of period $d = 3.93(4)$ μm and amplitude up to 80 nK on a two-dimensional ^{87}Rb gas, measure the sound speed along and across it, obtain $f_{s,x} = c_x^2/c_y^2$ and compare it with Leggett's formula on the imaged density [11]. Tao *et al.* excite phonons by Bragg scattering in a lattice of period $a = 266$ nm, up to 10 E_{R}, and find that "the speed of sound saturates Leggett's bound" [12]. Rabec *et al.* use a triangular lattice

of period $d = 6.0(1)\,\mu$m in a box of $42(1)\,\mu$m; a static method (density profile and continuity equation) and a dynamic one (compressibility and sound speed) agree and give $f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\,g\rho_0$ [24]. Biagioni *et al.* obtain f_s from the Josephson oscillations of single supersolid cells [25]. In uniform two-dimensional gases first and second sound give the superfluid density across the BKT transition [8]. In QMC the counterpart is the winding-number estimator [72], the response to a phase twist at the boundary [71].

Table 3.1: Dictionary from theory to measurement: the signal that stands for each quantity, and the main pitfall in reading one as the other.

Quantity	Measured signal	Pitfall
density $n(\mathbf{r})$	in-situ absorption [37]; fluorescence [46, 63]	column density; resolution
momenta $n(\mathbf{k})$	TOF image [2]; focusing in 2D [100]	far field ≈100 ms, not 20 ms
coherence $g_1(r)$	fringes [7, 104]; TOF peak width [23, 86]	sharp peaks above T_c [20]
superfluid fraction f_s	c_x^2/c_y^2 [11]; continuity eq. [24]; Josephson [25]	a tensor; needs two directions
sound speed c_s	density perturbation [40]; Bragg slope [4, 12]	two sounds in 2D [8]
$S(\mathbf{q},\omega)$	Bragg transfer [3, 61]	trap shift; beyond Bogoliubov [62]
gap, Higgs mode	modulation spectroscopy [45, 47]	heating read through a model
critical velocity v_c	heating behind a moving obstacle [5, 6]	vortices; instability [43]
compressibility κ	in-situ $n(\mu)$ with LDA [87]	LDA where density varies fast
number statistics	microscope [46, 63]; noise correlations [88]	occupation modulo two

What this means for theorists

A calculation is easiest to test when it produces the signal: the column density, or the density after a 20 ms expansion rather than the Fourier transform of the in-trap state [2]. For f_s the experiments report c_x, c_y and Leggett's formula on the measured density [11, 24]; a calculation that gives these three numbers can be checked directly.

Open

No Bragg measurement on a disordered or Bose-glass condensate was found in the literature checked for this map. Which signal of Table 3.1 separates a Bose glass from a weak superfluid in experiment is open: QMC uses winding numbers [26], the experiments coherence [13].

Key points

- Quasi-2D needs $\hbar\omega_z \gg \mu, k_BT$; experiments reach $\tilde{g} \sim 0.1$ [16, 37].
- A TOF image at 20 ms is not the momentum distribution of a coherent cloud [2].
- Bragg spectra give $S(\mathbf{q},\omega)$ and c_s; modulation spectra give gaps [4, 45].
- f_s comes from sound anisotropy, continuity or Josephson dynamics [11, 24, 25].

4 Data, benchmarks and verification

A benchmark is a number that a calculation without free parameters and an experiment both give for the same system; the list is short, and every entry comes with conditions.

4.1 Numbers that theory and experiment share

We call a comparison a benchmark when the theory has no adjustable parameter and the experiment measures the quantity the theory computes, not a proxy for it. Table 4.1 collects the pairs for which both numbers were checked for this map, for the Bose–Hubbard model, the two-dimensional (2D) Berezinskii–Kosterlitz–Thouless (BKT) transition and the superfluid fraction f_s.

The Bose–Hubbard critical points are the least clean benchmarks. In three dimensions the point measured by Mun *et al.* lies nearer the mean-field value than the quantum Monte Carlo (QMC) one [18], and Mark *et al.*, quoting the same two theory values, measure "number-dependent shifts in the excitation spectrum caused by effective multi-body interactions", which the standard model leaves out [91]. In two dimensions Spielman *et al.* locate the "superfluid to normal transition" where the condensed fraction vanishes, at 33(4) nK as reported, in agreement with QMC for a finite-sized system [110]; set against the ground-state value of an infinite lattice [19], the comparison crosses a temperature and a size.

Table 4.1: Benchmarks with a checked number on both sides. E_R is the recoil energy of the lattice in question (the quasicrystal papers write E_rec, others E_r). Theory is for homogeneous systems; the experiments are trapped or in a box, at finite temperature.

quantity	theory	experiment	source
$(U/J)_c$, 3D cubic, $\bar{n}=1$	QMC 29.34(2); mean field $5.8z = 34.8$	34.2(2.0) at 13.5(2) E_R (^{87}Rb)	[17, 18, 81, 91]
$(U/J)_c$, 2D square, $\bar{n}=1$	QMC $(J/U)_c = 0.05974(3)$, i.e. 16.74	condensed fraction vanishes at $U/t = 15.8(20)$ (as reported); microscope paper states 16.7	[19, 46, 110]
2D critical point	$D_c = \ln(C/\tilde{g})$, $C = 380 \pm 3$; μ_c constant $\tilde{\zeta}_\mu = 13.2 \pm 0.4$	trapped critical point matched "without any free parameters"; scale invariance near BKT	[15, 16, 58, 59, 92]
2D superfluid onset	$(\mu/k_BT)_c = 0.15$	frictionless flow from $\mu_\mathrm{loc}/k_BT \approx 0.24$	[6]
exponent at BKT	universal $\eta = 0.25$	$\eta_c = 0.17(3)$, trapped; $T_c/T_0 = 0.53(1)$	[38]
2D sound, box, $\tilde{g} = 0.16(1)$	Bogoliubov 1.6(1) mm/s; two-fluid 1.4(1) mm/s	1.49(3) mm/s at $T/T_c = 0.37(12)$	[40]
Bragg mean-field shift	$4n_0U/7h$ (local density)	$(0.54 \pm 0.07)\, n_0U/h$	[3]
f_s, 1D modulation of a 2D gas	Leggett bound is an identity; $f_{s,x} = c_x^2/c_y^2 = m/m_x^*$	sound anisotropy against Leggett's formula on the measured density	[11]
f_s, 1D lattice, $a = 266$ nm	Leggett's bound	"the speed of sound saturates Leggett's bound"	[12]
f_s tensor, triangular lattice	Gross–Pitaevskii; Leggett bounds of the measured profiles	$f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\, g\rho_0$; static and dynamic methods agree with both	[24]
superfluid–Bose-glass point, 8-fold quasicrystal	QMC $V_c/E_\mathrm{R} \simeq 1.76 \pm 0.01$ (weak interactions); $f_\mathrm{s} \approx 0$ in the glass	non-interacting localisation at 1.78(2) E_R	[13, 26, 52]
2D lattice criticality	$\mu_0 = -4t$; dilute Bose gas $z = 2$, $\nu = 1/2$	$\mu_0 = -4.5(6)t$; $z = 2.0(3)$, $\nu = 0.54(5)$	[99]

For the 2D gas, even at $mU \sim 0.01$ the mean-field n_s/n may not be used, because the fluctuation region is still of order T_c [92]. The two measurements that sit off the homogeneous predictions, the onset of frictionless flow [6] and the exponent at the transition [38], both compare a trapped gas with them. The 1.90(9) mm/s that Ville *et al.* obtain from standing waves comes from a separate experiment and is not comparable with the tabulated sound speed [40]. The three f_s experiments use three routes, all in the weakly interacting regime where the Gross–Pitaevskii equation (GPE) holds. In the quasicrystal, QMC and experiment agree on where the transition is but not through the same observable: QMC identifies the Bose glass by $f_s \approx 0$ from winding numbers while the compressibility crosses over smoothly [26], the experiments by the coherence of the momentum distribution [13].

4.2 Pitfalls

A time-of-flight (TOF) image of a coherent lattice gas is not its momentum distribution. The far-field time $t_{\mathrm{FF}} \approx m\, l_c R_0/\hbar$ is about 100 ms for $l_c \approx R_0 \approx 30d$ and $d \approx 400$ nm, against typical expansion times of 20 ms [2]. Sharp peaks are no proof of superfluidity either: QMC with up to 1.4×10^4 particles shows them above T_c, where the superfluid density and the condensate fraction both vanish [20]. Pollet *et al.* agree that peaks develop in the normal phase as the correlation length grows, but argue that a proper analysis of TOF images yields critical parameters accurately, whereas in-situ density profiles are "barely suitable for an accurate study of critical points in the strongly correlated regime" [111]. A benchmark calculation therefore has to model the image, not an ideal momentum distribution.

The visibility does not drop to zero at the Mott transition. Gerbier *et al.* report a contrast still around 30 % at 15 E_{R}, an insulator beyond 12.5 E_{R}, kinks near 14.1(8) and 16.6(9) E_{R}, and an exponent $\alpha = -0.98(7)$, consistent with a visibility proportional to $(U/zt)^{-1}$ (as reported) [86]; particle–hole theory reproduces this except at very large depths [22]. In one dimension, zero-temperature density-matrix renormalisation-group calculations show the transition "most directly in the half width of the interference peak" [23].

A critical velocity measured in a lattice need not be Landau's. Burger *et al.* lost superfluidity at $v_{\max} = 5.3(5)$ mm/s against a computed $c_{s,\max} \approx 5.2$ mm/s and read it as the onset of "the spontaneous emission of elementary excitations", without naming the Landau criterion [41]. In a moving lattice Fallani *et al.* found lifetimes falling from about 10 s to 10 ms at $q \approx 0.55\, q_B$, "unambiguously attributed to the onset of dynamical instability" after comparison with Gross–Pitaevskii theory [43]; currents in a very weak lattice become unstable above 0.5 recoil momentum [18]. Landau's criterion is a necessary condition for metastable superflow [68]; the moving-lattice threshold is set by the dynamical instability of the GPE instead.

Bragg spectra of lattice superfluids leave Bogoliubov theory behind. In a three-dimensional lattice the resonant frequency falls with depth faster than the Bogoliubov calculation, most strongly at 3.5 E_{R}, which Du *et al.* tentatively attribute to quantum depletion [62]. One-dimensional lattice superfluids show a continuum that "calls for an interpretation beyond the Bogoliubov spectrum" [45], and momentum-resolved Bragg data mark the limits of the theory [61].

Numerical references have their own ranges: the Gross–Pitaevskii-based truncated Wigner approximation (TWA) is "not suitable to formulate those fluctuations in the strongly interacting limit" [112], and the SU(3) discrete TWA built for that limit "showed noticeable deviations from the experimental results even at relatively short times" in a two-dimensional quench, according to a preprint [113].

Key points

- Critical-point comparisons set trapped, finite-temperature experiments against homogeneous ground-state QMC; in three dimensions the measured point lies nearer mean field.
- TOF images of coherent lattice gases are not momentum distributions and show sharp peaks above T_c; in lattices a measured critical velocity can be a dynamical instability, and Bragg spectra depart from Bogoliubov theory.
- Every measured f_s is in the weakly interacting regime, where it agrees with the GPE and with Leggett's bound.

5 State of the art

Each experimental line measured coherence first and superfluid response later; measurements of the superfluid fraction in modulated gases are recent and come from three groups.

The experiments fall into three lines: two-dimensional gases, lattice superfluids, and disordered or quasi-periodic lattices. Figure 5.1 sorts them by observable, Fig. 5.2 places them in time and Table 5.1 lists the groups. We write E_R for the recoil energy of the lattice in question; the papers write E_r, E_R or E_{rec}.

5.1 Two-dimensional gases

The two-dimensional (2D) line began with coherence. Hadzibabic *et al.* observed the Berezinskii–Kosterlitz–Thouless (BKT) crossover in a trapped ^{87}Rb gas by matter-wave heterodyning: "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. Cladé *et al.* produced a bimodal gas that "is quasi-condensed but is not superfluid" [21], and Tung *et al.*, imaging momentum space by 2D focusing, saw the compressibility ratio κ/κ_{ba} grow to 1.7 before a spike at $p = 0$ appeared [100]. Thermodynamics followed: scale invariance and "a wide universal region near the BKT phase transition" in caesium [58], and an equation of state at $\tilde{g} \approx 0.1$ in which phase-space density and entropy depend on μ/k_BT only, with entropies per particle down to $0.06\,k_B$ [37]. Sunami *et al.* made two planes interfere in a vertical double well to measure the correlation exponent at the transition [38].

Response measurements came with uniform traps. A laser obstacle stirred through a 2D gas gave $v_c/c_s = 0.3$–0.6 [6]. Box potentials of 200–900 μm^2 allowed quenches after which the mean number of topological defects follows the Kibble–Zurek scaling law [39], and an optical accordion compresses a single plane to frequencies above $\omega_z/2\pi = 10$ kHz [94]. In a box Ville *et al.* measured the sound speed and, above T_c, still saw strongly damped density waves [40]; theory treats collisionless sound [115], second sound [116] and sound across the transition [117]. Christodoulou *et al.* inferred from first and second sound a superfluid density that agrees with BKT theory, "including the prediction of a universal jump in the superfluid density at the critical temperature" (no sound speeds or T_c were read for this map) [8]. Fletcher *et al.* showed the BKT transition converging onto Bose–Einstein condensation as interactions vanish [59].

The superfluid fraction f_s of a modulated 2D gas has been measured since 2023: under a one-dimensional modulation of period 3.93(4) μm, by comparing the sound anisotropy with Leggett's formula on the measured density [11], and in a triangular lattice of period 6.0(1) μm at depths up to 4.7 μ_0, where static and dynamic methods and the Leggett bounds agree and the tensor is "nearly isotropic" [24]. Perrin's lecture notes, posted as a preprint, cover BKT physics

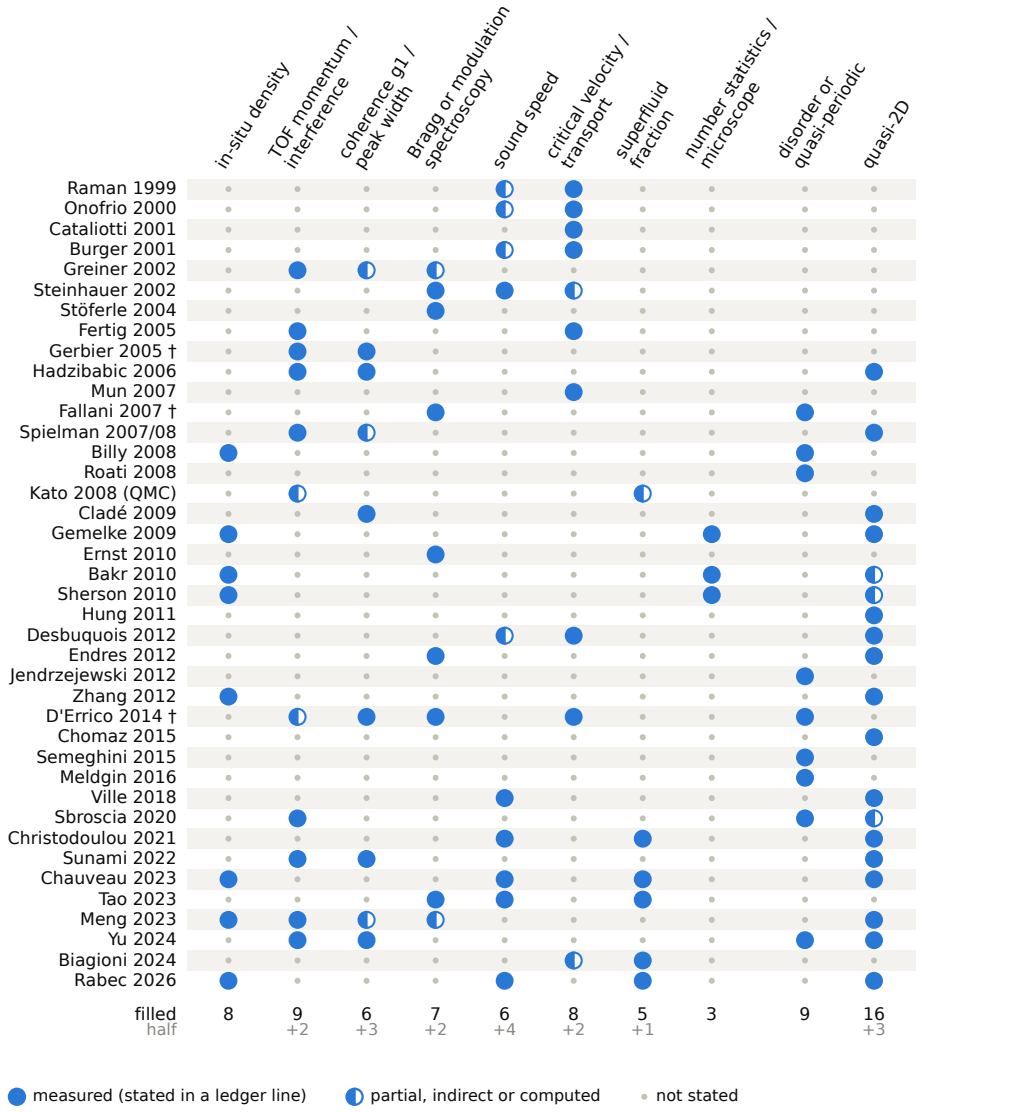


Figure 5.1: Coverage matrix: experiments of this map (rows) against observables (columns). Full dot: measured, as stated in a checked ledger line; half dot: partial, indirect or computed; small dot: not stated in the lines read, which does not mean never measured. Rows marked † rest on lines checked only in part.

and scaling symmetry [97].

5.2 Lattice superfluids

The superfluid–Mott-insulator (SF–MI) transition is still mostly characterised through coherence. Greiner *et al.* loaded up to 3000 sites of a 2D lattice and read the band populations by band mapping [101]; in the 3D lattice "at a potential depth of around 13 E_r the interference maxima no longer increase in strength … at a potential depth of 22 E_r no interference pattern is visible at all", with $U/J \approx 36$ at 13 E_R [31]. Noise correlations [88] and the residual visibility [22, 86] then probed the insulator. In two dimensions, momentum distributions agree with theory "(no adjustable parameters)" [36], and the condensate fraction falls from about 45 % in a shallow lattice to zero (as reported) [110]. In situ, Gemelke *et al.* measured the suppressed compressibility of Mott domains together with the temperature [87], Hung *et al.* the slow mass

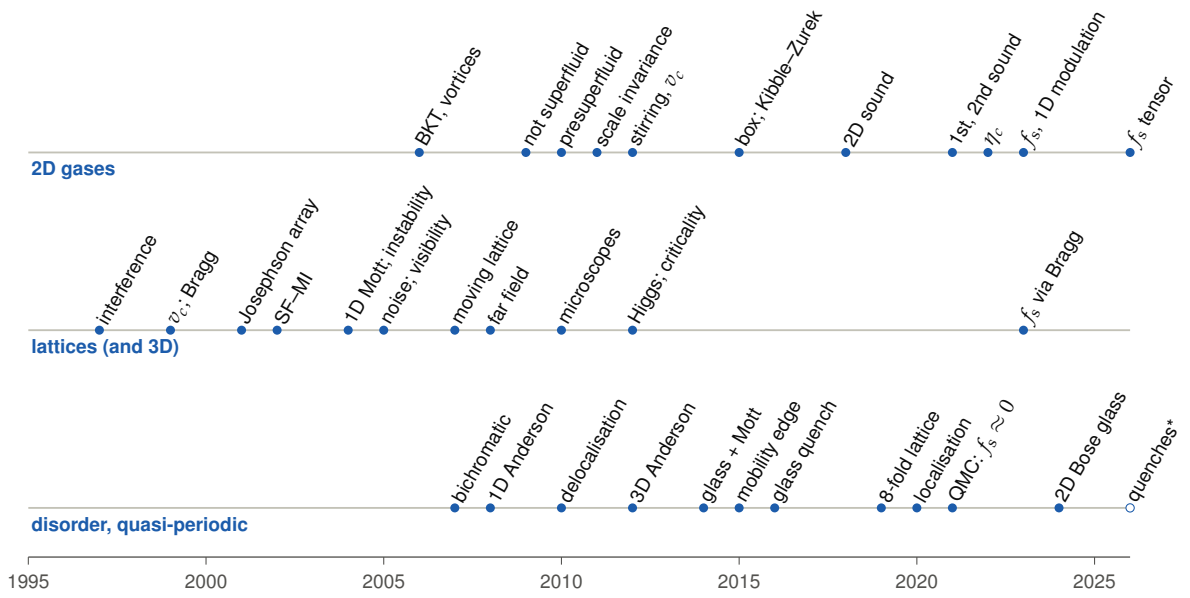


Figure 5.2: Timeline of the experiments discussed in this chapter, one dot per year and line (the 2021 quasicrystal dot is the QMC of Gautier *et al.*; open dot and asterisk: preprint). 2D gases: [6–8, 11, 21, 24, 38–40, 58, 59, 100]. Lattices and 3D: [2, 3, 5, 12, 18, 31, 42, 43, 45–47, 63, 86, 88, 99, 102]. Disorder and quasi-periodic: [9, 10, 13, 26, 48–53, 64, 109, 114].

transport across the transition [98], and Zhang *et al.* quantum-critical scaling [99]; quantum-gas microscopes resolved Mott shells up to $n = 4$ at $T/U \sim 0.16 \pm 0.03$ [46] and plateaus "with near-zero entropy" [63]. Trotzky *et al.* found the critical temperature suppressed near the Mott transition [118], and Mark *et al.* measured multi-body shifts [91]. Quenches across the transition gave power laws in the quench rate, $r = 0.31 \pm 0.03$ in a cubic lattice against the Kibble–Zurek $3/4$ [119], and coherence-length exponents well below the Kibble–Zurek value in one, two and three dimensions [120].

Transport probes the superfluid more directly. About 200 condensates coupled by tunnelling form a Josephson junction array [42]; the dipole frequency gives the band mass, $m^*/m = 1.18 \pm 0.02$ [41]; in one-dimensional gases damping is significant already at 0.25 E_R and the motion overdamped beyond 3 E_R [121]; moving lattices turn unstable [18, 43]; and a ring carries a persistent current whose critical velocity is consistent with vortex–antivortex pair creation [54].

Spectroscopy maps the excitations. Lattice modulation gives a broad continuum in the superfluid and discrete resonances in the Mott insulator [45], a response for which Kollath *et al.* gave the theory [106]. Bragg spectroscopy resolved band structure with indications of Bogoliubov backscattering [61], the drop of the response in the insulating regime [60] and deviations from Bogoliubov theory in three dimensions [62]; the excitations of condensates in one-dimensional periodic potentials have been measured [44] and computed [79, 80]. The amplitude mode was detected by Bragg spectroscopy [107], and Endres *et al.* saw its onset frequency soften towards the QMC critical point $j_c \simeq 0.06$ in lattices of 8–10 E_R [47]. Tao *et al.* closed the loop to f_s with Bragg phonons in a lattice of period 266 nm [12].

5.3 Disorder and quasi-periodic potentials

Anderson localisation of non-interacting matter waves came first: in a one-dimensional speckle beyond "an effective mobility edge", with algebraic decay otherwise [9]; in a bichromatic lattice, where "the critical disorder strength scales with the tunnelling energy" [10]; then in three dimensions [50], with a mobility edge measured "in the disorder–energy plane" [109]. A

Table 5.1: Groups whose experiments this map cites, named by senior authors on the papers' bylines; places are omitted because affiliations were not checked for this map. Lines: 2D gases (2D), lattices and three-dimensional gases (L), disorder and quasi-periodic (D), superfluid fraction (f_s).

group (senior authors)	lines	papers cited in this map
Dalibard, Beugnon	2D, f_s	[6, 7, 11, 24, 37, 39, 40, 94]
Hadzibabic	2D	[8, 59]
Chin	2D, L	[58, 87, 98, 99]
Cornell	2D	[100]
Foot	2D	[38, 95]
Phillips, Porto, Spielman, Campbell	L, 2D, f_s	[12, 21, 36, 54, 103, 110, 121]
Ketterle	L	[3, 5, 14, 18, 55, 56, 102]
Davidson (Steinhauer)	L	[4, 57]
Bloch	L, D	[2, 22, 31, 47, 63, 86, 88, 118, 120, 122]
Esslinger	L	[45, 101, 104]
Inguscio, Fort, Modugno	L, D, f_s	[10, 25, 41–44, 48, 49, 60, 64, 109, 123]
Aspect, Bouyer	D	[9, 50]
DeMarco	L, D	[65, 114, 119]
Schneider	D	[13, 51–53]
Sengstock; Greiner; Nägerl; Heinzen; Schneble	L, D	[46, 61, 62, 91, 107, 124]

Progress Article and a popular review summarise that period [32, 108].

With interactions the question becomes the Bose glass, "an insulating but compressible state without long-range phase coherence" [13, 30]. In a bichromatic lattice whose neighbouring sites differ in energy by up to 3.2 kHz $\simeq 1.7U$, the Mott resonances broaden into a flat density of excitations [48]. Repulsion delocalises a quasi-periodic gas through "an exponentially localized Anderson glass and the formation of locally coherent fragments as well as a coherent, extended state" [64] and makes transport subdiffusive [123]; at strong interaction and $k_B T \simeq 3J$ "a strongly-correlated Bose glass coexisting with a Mott insulator" appears [49]. Transport at unit filling shows a disorder-induced insulator "in quantitative agreement with a predicted superfluid–Bose-glass transition" [65], and disorder quenches give "evidence for superfluid puddles in the Bose glass phase" [114]. Glassy atomic mixtures [124], many-body localisation in two dimensions [122] and the theory of Bogoliubov excitations [125] and of superfluidity and sound [126] in disordered condensates extend the line.

Quasicrystals gave the 2D Bose glass. The first eightfold optical lattice produced self-similar diffraction [51], and Sbroscia *et al.* located the non-interacting localisation transition at 1.78(2) E_R from time-of-flight diffraction [52]. QMC placed the superfluid–Bose-glass point at $V_c/E_R \simeq 1.76 \pm 0.01$ for weak interactions and identified the glass by $f_s \approx 0$ from the winding-number estimator, while the compressibility crosses over smoothly [26]. Yu *et al.* then observed the Bose glass with the criterion "A high-contrast interference pattern composed of sharp peaks indicates the presence of long-range phase coherence, the signature of the superfluid phase. Localized states with only short-range coherence, however, result in an incoherent broad momentum distribution." The critical depth moves to deeper lattices as the scattering length grows, and "it is not possible to adiabatically traverse the Bose glass on typical experimental timescales" [13]. A preprint by the same group follows quenches across the transition, with timescales set by the quasi-disorder on the glass side and by tunnelling on the superfluid side [53]. Finite-temperature phases of trapped bosons have been computed with path-integral Monte Carlo [127], and a

preprint gives the thermodynamic phase diagram in a quasicrystal potential [128].

5.4 Who measures what

Interference and time-of-flight images appear in almost every line of Fig. 5.1. A measurement of f_s in a modulated gas, from sound, a density profile or a Josephson frequency, appears only in 2023–2026, from three groups [11, 12, 24, 25].

Key points

- 2D gases: coherence, thermodynamics and critical points are well measured; sound and the superfluid-density jump came in 2018–2021, and f_s under a modulation in 2023–2026.
- Lattices: the SF–MI transition is characterised by interference, in-situ densities and spectroscopy; direct superfluid responses are Josephson oscillations, damped transport and Bragg phonons.
- Disorder: localisation and the Bose glass are read from momentum widths, transport and spectra; the only $f_s \approx 0$ statement for a Bose glass comes from QMC.
- The 2D Bose glass cannot be crossed adiabatically on experimental timescales, which limits what a time-of-flight image can certify there.

6 Position: interference certifies coherence, not superfluidity

A sharp time-of-flight peak shows that phases are correlated over some distance; only a response to flow, a twist or a sound wave shows that the gas is superfluid.

Our claim covers weakly interacting bosons in periodic, modulated or disordered potentials and in two dimensions. It concerns what an observable certifies, not what the phases of these gases are. We argue it in four steps, the last applied to the twisted-bilayer optical lattice (TBOL); Figure 6.1 separates what is fixed from what is open, and Table 6.1 states what a response measurement would add, phase by phase.

What interference measures. Andrews *et al.* reported interference between two independent condensates [102]. The fringes record the relative phase of two matter waves, a first-order coherence property. Bloch, Hänsch and Esslinger measured the spatial correlation function directly with a radio-frequency double slit: above T_c it shows "a rapid gaussian decay, as expected for a thermal gas", below T_c "a slow decay towards a plateau" [104]. In two dimensions Hadzibabic *et al.* observed the Berezinskii–Kosterlitz–Thouless (BKT) crossover by matter-wave heterodyning, where "the loss of long-range coherence coincides with the onset of proliferation of free vortices" [7]. The measured quantity is the decay of coherence; the superfluid density enters through BKT theory. Fringe contrast of independent fluctuating condensates probes correlation functions [105]. In a lattice the superfluid shows sharp peaks, while in the Mott insulator "exact numbers of atoms are localized at individual lattice sites, with no phase coherence across the lattice" [31]. Each signal is a statement about phase correlations.

Where it misleads. Four results separate coherence from superfluidity. Quantum Monte Carlo (QMC) of the three-dimensional Bose–Hubbard model shows "that sharp peaks are not a reliable diagnostic of superfluidity": "even above the transition temperature Tc where both superfluid density and condensate fraction vanish, the interference pattern can nevertheless have sharp peaks riding over a broad background" [20]. A quasi-two-dimensional gas can be bimodal with

a coherence length longer than the thermal cloud's and still be "quasi-condensed but is not superfluid" [21]. A time-of-flight (TOF) image is not the momentum distribution unless the expansion reaches the far field; for phase-coherent samples "the far-field condition is usually not met for typical expansion times", with $t_{\mathrm{FF}} \approx 100\,\mathrm{ms}$ against the usual $t \approx 20\,\mathrm{ms}$ for a lattice spacing near 400 nm [2]. A Mott insulator still gives an interference pattern of finite visibility [22, 86], so a finite visibility does not by itself show a superfluid. The Bose-glass experiment in an optical quasicrystal states its criterion plainly: "A high-contrast interference pattern composed of sharp peaks indicates the presence of long-range phase coherence, the signature of the superfluid phase" [13]. This is a coherence criterion, and it works there because QMC confirms that the glass has $f_{\mathrm{s}} \approx 0$ (below). Pollet, Prokof'ev and Svistunov argue the other side: analysed properly, TOF images give critical parameters accurately, whereas in-situ profiles are poorly suited to locating critical points in the strongly correlated regime [111]. Both points hold: TOF can locate a transition when compared with a calculation of the same image, but by itself it does not certify superfluidity.

What certifies superfluidity. A superfluid is defined by how it responds; five responses are in use. (i) *Critical velocity.* A stirred condensate heated strongly only above a critical velocity of about 1.6 mm/s, almost a factor of four below the sound speed at the peak density [5]; a drag onset at $v_{\mathrm{c}} = (0.11 \pm 0.02)\, c_{\mathrm{s}}$ followed at the higher of two densities [14], and in two dimensions $v_{\mathrm{c}}/c_{\mathrm{s}} = 0.3$–$0.6$, with frictionless flow setting in at $\mu_{\mathrm{loc}}/k_B T \approx 0.24$ [6]. In a lattice, Burger *et al.* read the loss of superfluidity as the emission of elementary excitations [41], whereas Fallani *et al.* attributed the onset of dissipation to a dynamical instability [43]; this response therefore needs a model of the flow. (ii) *Moving frame and phase twist.* Leggett bounded the superfluid fraction by the density modulation [27] and later treated it for an arbitrary many-body system at $T = 0$ [69]. In the words of Pérez-Cruz, Astrakharchik and Massignan, "The superfluid fraction f of a quantum fluid is defined in terms of the response of the system to a weak and constant drag", with $E(v) = E(0) + f_{\mathrm{s}} N m v^2/2$ for $v \to 0$ and two-dimensional brackets $f^- \le f_{\mathrm{s}} \le f^+$ computed from the density alone [28]. The helicity modulus [71] expresses the same response through the boundary conditions. (iii) *Sound anisotropy.* Across a one-dimensional modulation $f_{s,x} = c_x^2/c_y^2$, and for a weakly interacting condensate "the bound reduces to an identity" [11]. Bragg phonons in a lattice showed that "the speed of sound saturates Leggett's bound" [12]. In a triangular lattice, $f_s^{(yy)} = 0.82(2)$ at $V_0 \simeq 2.4\, g\rho_0$, and the static and dynamic methods agree with each other, with Gross–Pitaevskii simulations and with the Leggett bounds of the measured profiles [24]. (iv) *Josephson oscillations.* In a supersolid, the Josephson oscillations of single cells gave "a relatively large sub-unity superfluid fraction" [25]; in an optical-lattice array, the square of the oscillation frequency gives the tunnelling rate and the Josephson critical current [42]. (v) *Winding numbers and second sound.* QMC samples the twist response through winding numbers [72]. In an eightfold quasicrystal it identifies the Bose glass by $f_{\mathrm{s}} \approx 0$ while the compressibility crosses over smoothly, since both phases are compressible [26]. First and second sound in a two-dimensional gas give a superfluid density that agrees with BKT theory, "including the prediction of a universal jump" [8].

The case. The one TBOL experiment with bosons found for this map builds its phase diagram from the visibilities of two families of TOF peaks, at the primary and at the moiré momenta [1]. The four phases are "superfluid, superfluid only with short-range coherence, Mott insulator, and insulator" (SF, SF-II, MI, I). SF shows both peak families; SF-II keeps the primary peaks and loses the moiré ones, and the Extended Data describe it as "superfluid domains embedded in a gapped insulator". Along path III, at a fixed depth of 8 or 14 E_{R}, "the visibility at the moiré momenta presents the threshold behavior and emerges at $\Omega_R > 0.5 E_r$" (the paper writes E_r for the recoil energy E_{R}). These are coherence statements. What is fixed: in SF, phases are correlated

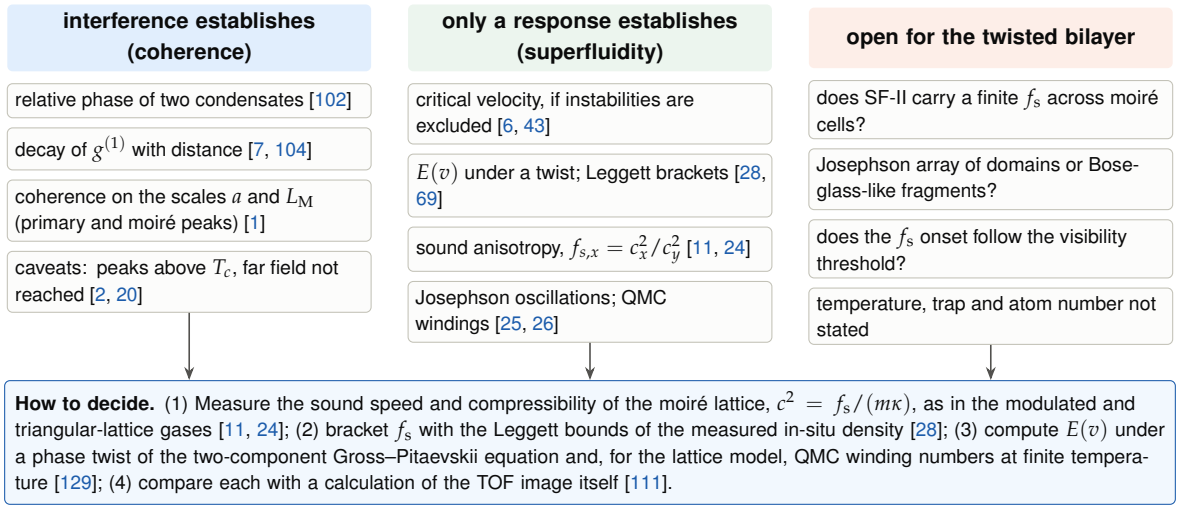


Figure 6.1: Position diagram. Left: what interference and TOF images establish; middle: the responses that certify superfluidity, each with its caveat; right: what stays open for the published TBOL phase diagram; bottom: the measurements and calculations that would decide.

over more than one moiré period, $L_M = 4.35\,\mu$m; in SF-II, over the primary lattice but not across moiré cells. What is open: whether SF-II carries a finite superfluid fraction on the scale of the sample, as a Josephson array of domains would; whether it is closer to a glass with $f_s \approx 0$, as in the quasicrystal [26]; and whether temperature has already depleted thin superfluid domains, as QMC finds in a single-component twisted potential [129]. The paper states no temperature, so the third reading cannot be excluded from the published data. Table 6.1 lists what a response measurement would add.

Table 6.1: What the visibility-based TBOL diagram says [1] and what a response measurement would say (outcomes that would be informative; none has been measured).

Phase	Visibility-based diagram	Response measurement
SF	both peak families: coherence beyond one moiré period	value of the f_s tensor, from sound and compressibility or a twist; how far below 1 (a triangular lattice gave $0.82(2)$ at $V_0 \simeq 2.4\,g\rho_0$ [24])
SF-II	primary peaks only: coherence within moiré cells, not across them	finite f_s with a soft inter-domain Josephson mode (array), or $f_s \approx 0$ with finite compressibility (glass-like) [26]
MI	no peaks	suppressed compressibility in situ [87]; a gap [31] and a discrete modulation spectrum [45]; $f_s = 0$
I	no peaks; filling modulated at the moiré scale	gapped or compressible: separates an insulator from a Bose glass [13, 30]
boundaries	threshold of a peak visibility, e.g. $\Omega_R > 0.5\,E_r$ on path III	onset of f_s along each axis, which at $T > 0$ need not coincide with the peak threshold [20]

Limits of the thesis. Three cases narrow the claim.

L1 TOF analysed against a full calculation of the image locates critical points [111], and ab-initio comparisons of experiment and QMC exist for lattice bosons [118]. The thesis concerns a visibility read without such a calculation.

L2 Where theory ties coherence to superfluid density, the two coincide: in a uniform two-dimensional gas the loss of coherence tracks vortex proliferation [7], the exponent at the transition was measured as $\eta_c = 0.17(3)$ [38], and sound confirms the universal jump [8].

Without a potential the question does not arise: at zero temperature a Galilean-invariant Bose fluid is expected to be fully superfluid [11].

L3 In a sample of a few moiré cells, coherence and f_s are both finite-size quantities; neither separates a phase from a crossover without a size dependence.

Where interference still earns its place

Band mapping keeps the band population [101], noise correlations reveal the order of a Mott insulator [88], the quasicrystal localisation point read from TOF, $1.78(2)\,E_{\mathrm{rec}}$ [13, 52], agrees with the QMC value 1.76 ± 0.01 [26], and the width of the interference peak tracks the transition in trapped one-dimensional systems [23]. The thesis asks only that a coherence signal be read as coherence.

Key points

- TOF peaks and visibilities measure first-order coherence; they can persist where $f_s = 0$ [20, 21].
- Superfluidity is certified by a response: critical velocity, twist, sound, Josephson oscillations or QMC windings; TOF compared with a full calculation can still locate transitions [111].
- The TBOL phases SF and SF-II are coherence statements; whether SF-II carries a finite f_s is open.

7 Around the topic

Supersolids, coupled condensates, flat-band lattices and photonic lattices ask the same question as a modulated condensate, which part of the fluid is superfluid, with one ingredient pushed to a limit.

7.1 Supersolids and their superfluid fraction

A supersolid is a condensate whose density is modulated by its own interactions instead of an external potential. For the superfluid fraction f_s this changes little: the density is periodic, the flow has to pass through its minima, and Leggett's argument applies unchanged [27]. Biagioni *et al.* measured f_s of a supersolid from the Josephson oscillations of single cells and found "a relatively large sub-unity superfluid fraction" [25]. It is a third experimental route to f_s, beside the sound speeds and the static density profile of Chapter 4.

Theory reached the same quantity from classical-field models. Josserand, Pomeau and Rica wrote a model of a defect-free supersolid in which ordinary elasticity and superfluidity coexist [130]; Sepúlveda, Josserand and Rica computed the nonclassical rotational inertia (NCRI) fraction in a one-dimensional version [131]; and Aftalion, Blanc and Jerrard state, in a preprint, "Using a model of quantum crystal introduced by Josserand, Pomeau and Rica, we prove that NCRI occurs" [132]. Blakie treated the two-dimensional supersolid in the Gross–Pitaevskii picture, where "The superfluid fraction is described by a rank-2 tensor" and nonclassical translational inertia is tied to the effective mass [70]: the same tensor that Rabec *et al.* measured in an external triangular lattice. Šindik *et al.* obtained f_s of a ring dipolar supersolid from its sound modes and compared it, as the preprint describes, with the nonclassical moment of inertia and with Leggett's estimate [133]. Two further preprints extend the method side: Pizzino *et al.* examine Leggett's bound for strongly interacting bosons at the crossover from two to one dimensions (the

conclusion was not checked for this map) [75], and another proposes quantum vortex necklaces as a probe of superfluidity [134].

7.2 Coupled condensates and Josephson physics

A deep lattice is an array of weak links. Cataliotti *et al.* made this literal with about 200 wells of about 1000 atoms each, in which "The square of the small-amplitude oscillation frequency is proportional to the microscopic tunneling rate" [42]; the superfluid current is then a Josephson current, and the stiffness is set by the tunnelling rate. The target system of Chapter 8 adds a second coupling, between internal states. For two coherently coupled condensates Abad and Recati showed that a finite coupling changes the first-order demixing transition into a second-order one [135]; the coupling also gaps the relative-phase branch of the spectrum, a mode with no analogue in a single condensate. Experiments have studied ferromagnetism in an extended coherently coupled superfluid [136] and observed "bubble nucleation in isolated and coherently coupled atomic superfluids", a false-vacuum decay [137].

Tunnel-coupled one-dimensional gases are Josephson systems of the same kind. In thermal equilibrium they act as "a quantum simulator of the sine-Gordon model" [138]; with Raman coupling they would show a commensurate–incommensurate transition "described by the Pokrovsky-Talapov model for the relative phase", with "a soliton staircase in the phase difference" [139]. Relaxation in an extended bosonic Josephson junction of this type was modelled by "stochastic classical field simulations using Gross-Pitaevskii equations in three spatial dimensions", which "reproduce the main experimental findings", and the relaxation persists "even at zero temperature" [140]; an earlier study by several of the same authors treated fluctuations and stochastic processes in one-dimensional many-body quantum systems [141].

7.3 Flat-band bosons: the limit $m/m^* \to 0$

For a weakly interacting condensate in a periodic potential, Chauveau *et al.* state that the superfluid fraction "exactly coincides with the ratio m/m_x^*" [11], m^* being the band mass that also sets the sound speed [73]. A flat band is the limit $m/m^* \to 0$ of the single-particle band mass, so any superfluidity there must come from interactions and from the geometry of the Bloch states. Julku, Bruun and Törmä showed for the kagome lattice that the sound velocity $c_s = (2Un_0/3)\sqrt{g(k_c)}$ is linear in U and that the depletion stays finite as $U \to 0$, set by the quantum geometry alone [142] (a longer account is in [143]); a later revision fixes the particle number when the superfluid weight is taken [144]. Huber and Altman had projected the interactions onto the flat band and found a Wigner-crystal threshold $\nu_c = 1/9$ on the kagome lattice [145]. Two 2026 preprints sharpen the conditions: "There can therefore be no superfluidity in a single flat band at a TRIM where the Bloch state has uniform densities" (TRIM: time-reversal-invariant momentum), and a stable two-dimensional flat-band superfluid "requires at least three bands within Bogoliubov theory" [146]; a second preprint relates condensation to the geometry of compact localised states [147]. Other preprints relate the sound speed to a quantum geometric kernel [148] and find that for a two-band condensate "in the case of U(2) invariant interactions, the superfluid density is zero" [149].

Experiments with bosons in flat-band lattices exist, but none of those read for this map measures f_s. In a kagome lattice, ^{87}Rb condensates show an interaction-enhanced group velocity in the flat band that agrees "quantitatively with predictions of the lattice Gross-Pitaevskii equation" [150]; bosons have also been loaded into Lieb [151] and tunable kagome [152] lattices, and topological Bogoliubov quasiparticles have been studied for a condensate in a flat-band system [153].

7.4 Photonic analogues

Paraxial light in a nonlinear medium obeys a Schrödinger equation with a cubic term, so a photonic lattice is a mean-field model of a condensate without temperature or a superfluid fraction. Photonic moiré lattices show the two-dimensional localisation–delocalisation transition of light [154], and the formation of solitons controlled by the twist angle has "threshold properties that are a pristine direct manifestation of flat-band physics" [155]. These experiments test the single-particle and nonlinear-wave parts of the twisted-lattice problem, which Chapter 8 needs, but not its superfluid response.

Key points

- Supersolids add a third measured route to f_s, Josephson oscillations of single cells, and share the tensor description with external lattices.
- Deep lattices, coupled one-dimensional gases and coherently coupled condensates are Josephson systems: the superfluid current is a tunnelling current, and a coherent coupling gaps the relative-phase mode.
- A flat band is the limit $m/m^* \to 0$: superfluidity there rests on interactions and band geometry, and within Bogoliubov theory a stable two-dimensional flat-band superfluid needs at least three bands (preprint).
- No flat-band or photonic experiment has measured a superfluid fraction in the sources read for this map.

8 Observables for the twisted-bilayer problem

Each observable of the earlier chapters set against the one twisted-bilayer experiment with bosons: what it measured, what it left out, and what each reference measurement would add.

This chapter describes and does not plan. Table 8.1 lists, for each observable, whether the twisted-bilayer optical lattice (TBOL) experiment measured it, what it certifies (Chapter 6), its template experiment and the calculation that would predict it.

8.1 What the experiment measured and what it left out

Meng *et al.* load ^{87}Rb into two square lattices of constant $a = 395\,\mathrm{nm}$, each felt by one hyperfine state, $|1\rangle = |F{=}1, m_F{=}1\rangle$ or $|2\rangle = |F{=}2, m_F{=}0\rangle$, and twist them by $\theta = 5.21°$; the moiré period is $L_\mathrm{M} = 4.35\,\mu\mathrm{m}$, eleven lattice constants [1]. The lattices use tune-out wavelengths, characterised by Kapitza–Dirac scattering in a preprint [156]. A 6.8 GHz microwave couples the layers with a Rabi frequency Ω up to about $1\,E_\mathrm{R}$, with $E_\mathrm{R} = h \times 3.67\,\mathrm{kHz}$ (written E_r in the paper). An accordion lattice, tunable from 12 to 3 μm, holds the gas in a pancake with "$\omega_z \approx 2\pi \times 1\,\mathrm{kHz}$". The data are microwave spectra at $V_0 = 0$, 4 and $10\,E_\mathrm{R}$, in-situ images at 50× magnification that resolve the moiré density pattern, and state-selective momentum distributions after 18 ms of time of flight (TOF). At $4\,E_\mathrm{R}$ the calibrated coupling is $2\pi \times 893\,\mathrm{Hz} = 0.243\,E_\mathrm{R}$, or $\Omega_R/t = 2.07$, and the paper quotes $U/t = 1.67$ (its tie to that depth was not seen in the checks). Deepening the lattice at $\Omega_R = 0.24\,E_\mathrm{R}$ (path I) takes the gas from the superfluid (SF) through SF-II, "superfluid only with short-range coherence", to the Mott insulator (MI); raising Ω at 8 or $14\,E_\mathrm{R}$ (path III) reverses the sequence, with the moiré visibility emerging at $\Omega_R > 0.5\,E_r$. The theory in the paper is a two-component Gross–Pitaevskii equation (GPE) with "$g = g_{11} = g_{22} = g_{12}$" and a single-site Gutzwiller treatment of the Bose–Hubbard model.

Four inputs that every observable below depends on were not found in the paper, as read for this map: the temperature, the radial trap frequency, the atom number in the pancake (the condensate is made with up to 5×10^5 atoms) and scattering lengths for the pair $|1,1\rangle, |2,0\rangle$. Measured values exist for other pairs, $a_{12} = 98.006(16)\, a_0$ and $a_{22} = 95.44(7)\, a_0$ for $|1,-1\rangle, |2,+1\rangle$ [157], and an earlier scattering-length set gives, as reported, a marginally miscible mixture [158]. The data statement reads "All data generated or analysed during this study are included in this published article." [1].

Table 8.1: Observables for the TBOL problem. "Measured" refers to Meng *et al.* [1] as read for this map; the last column names the calculation that would predict the observable, not a plan.

Observable	Measured in TBOL?	What it certifies	Template experiment	Calculation to compare
in-situ density	yes, L_M resolved	density modulation; input to Leggett brackets	[11, 24]	GPE ground state; brackets [28]
local compressibility	no	incompressible (Mott) vs compressible	[87]	QMC, GPE equation of state
primary TOF peaks	yes, visibility	coherence on the scale a	[31, 86]	TOF of the trapped state, near field [2]
moiré TOF peaks	yes; threshold $\Omega_R > 0.5\, E_r$	coherence beyond L_M	[13]	same, at finite T [20]
microwave spectra	yes, $V_0 = 0$–$10\, E_R$	single-particle bands, calibration	—	band structure
sound speed, κ	no	f_s tensor, $c^2 = f_s/(m\kappa)$	[12, 24]	Bogoliubov–de Gennes sound
phase twist, moving lattice	no	f_s tensor, $E(v)$	[11]	twisted GPE [28, 69]
Josephson oscillations	no	inter-domain coupling; f_s	[25, 42]	GPE plasma mode
Bragg, modulation	no	gaps, Bogoliubov bands	[45, 61]	Bogoliubov–de Gennes, $S(q,\omega)$
critical velocity	no	superflow below v_c; needs a stability model	[6, 43]	time-dependent GPE
winding numbers	— (theory)	f_s	—	QMC [26, 129]
temperature	not stated	needed by every row	[24, 63]	—

8.2 Coherence observables

In-situ density. The images resolve the 4.35 μm pattern; whether the 395 nm lattice itself is resolved was not found in the sources checked. The Leggett brackets need such a map, and the two sound-based f_s experiments in two-dimensional gases computed them from measured profiles [11, 24]. The two-dimensional brackets [28] require $n(\mathbf{r})$ on the scale of every modulation. Because $1/n$ is convex, a density averaged over the unresolved lattice still gives a valid but looser upper bound f^+, while the lower bound f^- loses its guarantee (our remark). In-situ images also give the local compressibility, suppressed in a Mott domain [87].

Primary and moiré peaks. The moiré peaks sit at one eleventh of the primary spacing, since $L_M = 11a$. Their visibilities fix phase coherence on the scales a and L_M, and nothing more (Chapter 6). Whether 18 ms reaches the far field cannot be judged, since the far-field time grows with sample size and coherence length and the radial size is not stated; for a 400 nm lattice with both lengths near 30 sites, $t_{FF} \approx 100$ ms [2]. The criterion used for the quasicrystal Bose glass reads sharp peaks as long-range coherence and a broad distribution as localised states [13];

applied here, SF-II has coherence over the primary lattice and none detected across moiré cells.

Microwave spectroscopy. The spectra calibrate the coupling [1]; they probe single-particle transitions, not the interacting gas.

8.3 Response observables and what each would add

Sound and phase twist. The moiré period, 4.35 μm, lies between the two modulations on which the superfluid fraction has been measured through sound: 3.93(4) μm [11] and 6.0(1) μm [24]. The triangular-lattice experiment is the template. It used a box of $L = 42(1)\,\mu$m, $\rho_0 = 51(2)\,\mu\text{m}^{-2}$, $g\rho_0/k_B = 45(2)\,$nK, $T < 20\,$nK and $\omega_z/2\pi = 4.1(2)\,$kHz, and obtained the tensor statically, from the density profile and the continuity equation, and dynamically, from the compressibility and the sound speed [24]. The measured tensor was nearly isotropic [24]. For a moiré pattern that keeps the fourfold symmetry of the square lattices, the tensor is isotropic by symmetry (our remark), so the ratio c_x^2/c_y^2 [11] carries no information and the dynamic route through $c^2 = f_s/(m\kappa)$ is the one that transfers. Compared with the template, the TBOL gas has two coupled components, a 395 nm lattice under the moiré pattern, an in-plane trap that the paper does not specify, and an axial frequency about four times lower. A phase twist of the two-component GPE, $E(v) = E(0) + f_s N m v^2/2$, gives the same tensor in a calculation [28].

Bragg and modulation spectroscopy. Momentum-resolved Bragg spectroscopy maps the band structure of a lattice superfluid and shows where Bogoliubov theory applies and where it fails [61]; in a three-dimensional lattice the resonance at fixed momentum falls with depth faster than Bogoliubov theory predicts [62]. Modulation spectroscopy shows a broad continuum in the superfluid and a discrete spectrum in the Mott insulator [45], and the Bragg response drops in the insulating regime [60]. For the TBOL these would test the "gapped insulator" around the SF-II domains and resolve the Bogoliubov bands of the moiré cell, including the relative-phase branch of a coherently coupled condensate [135].

The moiré Bose–Hubbard mapping. In two dimensions the square-lattice benchmark is $(J/U)_c = 0.05974(3)$, i.e. $(U/J)_c = 16.74(1)$, from quantum Monte Carlo (QMC) [19]; mean field gives $5.8z$ [81], and cluster Gutzwiller extrapolated from clusters of up to 4×4 sites reaches $0.0596(4)$ [159]. Its limits are known: in hexagonal potentials "the Bose–Hubbard approximation fails to describe but the first Mott lobe" at $15\,E_R$ [89], and multi-body terms appear in the spectrum [91]. For twisted bilayers, Paul, Recher and Santos use Gaussian-overlap interlayer hopping, merge strongly coupled site pairs into clusters above $t_{\perp,\text{cr}} = 0.03\,t_\perp$ (commensurate) or $0.06\,t_\perp$ (incommensurate), and find "mobility islands" [160]; incommensurate twists act as effective disorder with multifractal states [161]. Without interlayer tunnelling, contact interactions alone give interlocked Mott phases [162]. For the experiment's single-site Gutzwiller treatment of the moiré case, no benchmark of this kind was found for this map.

The Josephson-array reading of SF-II. If SF-II is an array of coherent domains coupled by tunnelling, its superfluid fraction is set, on this reading, by the inter-domain band mass: in the modulated gas the superfluid fraction "exactly coincides with the ratio m/m_x^*" [11], and the sound speed follows the Taylor–Zaremba form with the band mass [73]. Josephson observables test it: an array's oscillation frequency gives the tunnelling rate [42], single cells give f_s [25].

The Bose-glass alternative. A Bose glass is "an insulating but compressible state without long-range phase coherence" [13]. QMC identifies it by $f_s \approx 0$ from winding numbers while the compressibility changes smoothly [26], and a disorder quench found "superfluid puddles in the Bose glass phase" [114]. A lattice study of the twisted system finds that stronger interactions turn a disconnected Bose glass into a connected superfluid at low filling [163]. Separating the

readings takes compressibility and f_s together.

Quasi-two-dimensional conditions and temperature. Freezing the axial motion needs $\hbar\omega_z \gg k_B T$ and $\hbar\omega_z \gg gn$ [16, 33]. The TBOL pancake has $\omega_z \approx 2\pi \times 1\,\mathrm{kHz}$, against $1.9(2)\,\mathrm{kHz}$ in the Paris equation-of-state experiment [37] and $4.1(2)\,\mathrm{kHz}$ in the triangular lattice [24]; without T and n the two ratios cannot be formed. For thin domains temperature decides: QMC of a single-component twisted potential at $k_B T = 0.01$–$0.04\,E_R$ finds maximal hoppings of 10^{-11}–$10^{-3}\,E_R$ in the superfluid domains, and "thermal broadening strongly depletes the SF domains" [129].

Calculations to compare with. Simulations must assume what the experiment does not state: a Floquet GPE study takes $N = 2 \times 10^4$ and a trap of $2\pi \times (20, 20, 1000)\,\mathrm{Hz}$ at $V_0 = 4\,E_R$, $\Omega = E_R$, $\theta = 9.43°$ [164] (preprint). A uniform Ω maps onto a scalar moiré potential in three dimensions [165] (preprint) and, with detuning, onto a Hofstadter model [166]; Kariyado–Vishwanath flat bands need a spatially varying coupling [167]. The potentials were proposed by González-Tudela and Cirac [168].

Key points

- The TBOL experiment measured coherence, the in-situ moiré density and microwave spectra.
- Temperature, radial trap, pancake atom number and scattering lengths were not found in the paper.
- The f_s template is the triangular-lattice sound measurement, period $6.0(1)\,\mu\mathrm{m}$.
- Josephson and compressibility data would separate a domain array from a glass.

9 Open questions

Twenty-two questions in five themes, from the meaning of the Landau criterion in a lattice to the superfluid fraction of the twisted bilayer, followed by directions that no checked study has taken up.

The questions follow from Chapters 6 and 8, grouped in Figure 9.1; "not found" refers to the literature checked for this map (cutoff 23 September 2026).

definitions, Landau	**Q1** Landau or instability	**Q2** why $v_c/c_s < 1$	**Q3** which f_s at $T > 0$	**Q4** Leggett beyond 1D	
measuring f_s	**Q5** deep lattices	**Q6** sound near BKT	**Q7** brackets in disorder	**Q8** Josephson route	**Q9** new probes
Bose–Hubbard observables	**Q10** visibility against f_s	**Q11** critical points in traps	**Q12** beyond one band	**Q13** gapped modes	**Q14** methods at finite T
disorder, localisation	**Q15** f_s of a Bose glass	**Q16** disorder excitations	**Q17** mobility edge	**Q18** quasicrystal dynamics	
twisted bilayer	**Q19** is SF-II superfluid?	**Q20** effective model	**Q21** extended condensate	**Q22** T, two components	

Figure 9.1: The 22 open questions, one row per theme; the rows follow the subsections below.

9.1 Definitions and the Landau criterion

Q1 Landau or dynamical instability. In moving lattices dissipation was read as the emission of elementary excitations [41], as a dynamical instability [43] and as current instability

above half a recoil momentum [18]. Which criterion sets v_c in a two-dimensional or moiré lattice?

Q2 **Why $v_c/c_s < 1$.** Measured ratios lie between 0.07 and 0.6 [5, 6, 14], and a persistent current stops in a way consistent with vortex–antivortex creation [54]. Landau's criterion is only necessary [66–68]; what sets the ratio for an obstacle as large as the lattice period is open.

Q3 **Which f_s at $T > 0$.** The moving-frame fraction [69], the helicity modulus [71], winding numbers [72] and the Berezinskii–Kosterlitz–Thouless (BKT) jump $n_s\lambda^2 = 4$ [16, 34, 35, 93] are distinct definitions; no comparison for a modulated gas, where f_s is a tensor [70], was found.

Q4 **Leggett beyond one dimension.** The bound is an identity for a one-dimensional modulation [11], saturated in a lattice [12] and respected in a triangular lattice [24]. Angular dependence [74], strong interactions [75] and supersolid models [130–132] are the next tests ([74, 75, 132] are preprints).

9.2 Measuring the superfluid fraction in lattices and two dimensions

Q5 **Deep lattices.** Sound gave the tensor up to $V_0 \simeq 4.7\,\mu_0$ [24], Bragg phonons up to $10\,E_R$ [12]. Does sound still give f_s where the lattice Bogoliubov bands [44, 79, 80] give way to a continuum near the Mott transition [45]?

Q6 **Sound near BKT with a lattice.** In a box $c = 1.49(3)$ mm/s at $T/T_c = 0.37(12)$ against a Bogoliubov 1.6(1) mm/s [40, 115], and sound gives the superfluid-density jump [8, 116, 117]. How a modulation changes each tensor component is open.

Q7 **Leggett brackets in disorder.** Sound and f_s were computed for disordered gases [126], and Bogoliubov quasiparticles in a preprint [125]; brackets from a measured speckle density were not found.

Q8 **The Josephson route.** Single supersolid cells give f_s [25]; a lattice array gave the tunnelling rate [42], and coupled one-dimensional gases realise the sine-Gordon model [138]. Josephson f_s in a two-dimensional lattice is open.

Q9 **New probes.** Vortex necklaces [134] and supersolid sound [133] (preprints), noise correlations [88] and single-site snapshots [46, 63]: whether they constrain the stiffness in a lattice is open.

9.3 Bose–Hubbard observables

Q10 **Visibility against f_s.** Time-of-flight (TOF) condensate fractions [36, 110] and residual visibilities [22, 86] are measured, peaks survive above T_c [20], yet no quantitative visibility–f_s relation at experimental temperatures was found.

Q11 **Critical points in traps.** The measured three-dimensional $(U/J)_c = 34.2(2.0)$ lies 2σ from the quantum Monte Carlo (QMC) 29.34(2) [17, 18]; QMC gives 16.74(1) in two dimensions [19], density-matrix renormalisation $(U/t)_c \approx 3.61$ in one [82], after early QMC [83]; criticality was seen in situ [99]. Where the f_s onset sits against the coherence onset at $T > 0$ [111, 118] is open.

Q12 **Beyond one band.** Density-assisted tunnelling [90], multi-body shifts [91] and continuous-space QMC [89] go beyond the Bose–Hubbard model [29, 30, 84, 85]; which observable feels this most is open.

Q13 **Gapped modes.** The amplitude mode softens towards $j_c \approx 0.06$ [47] and is seen in Bragg spectra [107], with modulation theory in [106]. How gapped modes constrain f_s is open.

Q14 **Methods at $T > 0$.** Ramps give an exponent 0.31 against the Kibble–Zurek 0.75 [119] and depend on the final interaction [120]; transport is slow [98]. Worm QMC [169], tensor networks [170, 171], neural states [172] and classical fields [141, 173–176] compete; truncated Wigner is "not suitable to formulate those fluctuations in the strongly interacting limit" [112], a cluster variant deviates from experiment [113] (preprint), and a spin-1 study takes, as reported, the superfluid tensor from momentum fluctuations [177]. Which method gives f_s of a modulated two-dimensional gas is open.

9.4 Disorder and localised excitations

Q15 **f_s of a Bose glass.** QMC finds $f_s \approx 0$ [26]; experiments see the glass through coherence [13], spectra [48], transport [65], quenches [114], a second species [124] and the disorder–interaction plane [49]. No direct f_s measurement was found.

Q16 **Excitations in disorder.** Bogoliubov quasiparticles in disorder [125] and a "flat density of excitations" [48] are the starting points; no Bragg experiment on a disordered condensate was found for this map.

Q17 **Interactions and the mobility edge.** Localisation [9, 10, 50] and a mobility edge [109] are measured, interactions give "locally coherent fragments" [64] and subdiffusion [123] (see [32, 108]); what f_s does is open.

Q18 **Quasicrystals at $T > 0$ and out of equilibrium.** The glass cannot be traversed adiabatically on typical experimental timescales [13]. Across quantum walks [53], finite-T diagrams [127, 128] ([53, 128]: preprints), many-body localisation [122], eightfold [51, 52] and rotated lattices [178], no response measurement was found.

9.5 The twisted-bilayer target

Q19 **Is SF-II superfluid?** "Superfluid only with short-range coherence" is a coherence label [1]. Theory finds "mobility islands" [160], interlocked Mott phases [162], insulating islands in a superfluid [179] and a "local supersolid" [180] (preprint).

Q20 **Which effective model.** The scalar potential [165], the Hofstadter limit [166], the cluster picture [160, 161], fitted tight binding [181] and the proposal [168] are not joined; uniform coupling gives no Kariyado–Vishwanath flat band [167], unlike honeycomb bilayers [182–184].

Q21 **The extended condensate.** Interactions broaden moiré bands [185], self-organise supercells [186] and leave vortex pairs [187] (preprints); gap-soliton [188–192], photonic [154, 155] and Floquet [164] abstracts give no Bogoliubov spectra.

Q22 **Temperature and two components.** When thin domains are depleted [129], and how the relative-phase branch [135], miscibility [157, 158], magnetism [136] and false-vacuum decay [137] of coupled condensates enter a moiré cell, is open; coupled one-dimensional gases give a junction limit [139, 140].

9.6 Directions not yet systematically addressed

As far as the checks for this map could establish, no study has taken up the following.

D1 An f_s measurement on a moiré or Rabi-coupled gas (the four so far: [11, 12, 24, 25]).

D2 A direct f_s measurement, or Bragg spectroscopy, of a Bose glass.

D3 Visibility and f_s computed for one trapped, modulated two-dimensional gas at the experiment's expansion time [2].

D4 Bogoliubov spectra, sound and the f_s tensor of the Rabi-coupled twisted condensate.

D5 Finite-temperature or truncated-Wigner studies of the Rabi-coupled model.

D6 Ramps or quenches of the coupling (other quenches: [163, 164, 187]).

D7 Leggett brackets from the measured density of a potential with two length scales.

Key points

- Which instability sets v_c in a lattice, and why $v_c/c_s < 1$, is not settled.
- The superfluid fraction has been measured in four modulated systems, none moiré and none disordered.
- No quantitative link between TOF visibility and f_s at experimental temperatures was found.
- For the twisted bilayer, a response measurement and an effective model are both missing.

10 Reading paths and outlook

Where to start reading, depending on the background one brings, and what the next measurements and calculations would settle.

10.1 Where to start

Theorists new to the laboratory should read chapter 3, then the review of two-dimensional gases by Hadzibabic and Dalibard [16], the review of Bloch, Dalibard and Zwerger [84] and two sets of lecture notes, Dalibard's course on coherence and superfluidity [68] and Perrin's course on two-dimensional gases, posted as a preprint [97]. What an image can show is set by the far-field analysis of Gerbier *et al.* [2] and by the QMC finding that sharp peaks survive above T_c [20]; Pitaevskii and Stringari's monograph is the theory reference [67].

Experimentalists who want the definitions should start from Leggett [27, 66, 69], then the drag definition and the two-dimensional brackets of Pérez-Cruz *et al.* [28] and the sound measurement of Chauveau *et al.* [11]. The helicity modulus [71], the winding-number estimator [72] and the universal jump [93] are what a QMC paper reports.

Numerical analysts find the models in chapter 2 and the checks in chapter 4. Gross–Pitaevskii and Bogoliubov calculations [76, 77] are tested against the Bragg data [3, 4] and the lattice Bogoliubov bands and sound speeds [73, 79, 80]. Bose–Hubbard codes are tested against the QMC and DMRG critical points [17, 19, 82] and the cluster-Gutzwiller extrapolation [159]; an open worm-algorithm code exists [169], and Krutitsky reviews the lattice results [85]. Two-dimensional simulations are tested against the classical-field BKT constants [15, 92], and the validity conditions of classical-field and truncated-Wigner methods are set out in [173–176].

10.2 Outlook

The superfluid fraction of a modulated condensate is now measured, through sound in two laboratories and through Josephson oscillations in a third [11, 12, 24, 25], all in condensates away from an insulator. The observables of this map can disagree further out: near the Mott transition, where the critical temperature is suppressed [118]; in the Bose glass, where QMC gives $f_s \approx 0$ [26] and no measurement of f_s was found for this map; and near the BKT point, where the measured coherence exponent departs from its universal value [38]. A response measurement in any of these, set against a calculation of the same quantity, would test the dictionary proposed here; chapter 8 does this for the twisted-bilayer lattice.

Key points

- Theorists: the toolbox chapter, the reviews and lecture notes, then what an image shows.
- Experimentalists: Leggett, Pérez-Cruz *et al.* and Chauveau *et al.*; numerical analysts: the Bragg, QMC and BKT benchmarks.
- Next: superfluid-fraction measurements where coherence and superfluidity part ways.

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A Glossary

Terms from superfluidity, two-dimensional gases, optical lattices, disorder and cold-atom measurement, as this map uses them.

Landau criterion. A fluid flowing past a wall or obstacle at velocity v can create an excitation of momentum p and energy $\varepsilon(p)$ only if $v > \varepsilon(p)/p$; below the minimum of this ratio the flow cannot decay by emitting excitations. A necessary condition for metastable superflow.

Critical velocity. The velocity v_c above which a moving obstacle or a flow starts to dissipate. For a Bogoliubov spectrum Landau's value is the sound speed; measured values lie well below it, and vortex nucleation is one reason.

Superfluid fraction. The fraction f_s of the mass that does not follow a slow motion of the walls or of the potential. In an anisotropic potential it is a tensor.

Non-classical rotational or translational inertia. The reduction of the moment of inertia (or of the translational mass) below its classical value, the operational definition of f_s in a rotating or moving frame.

Leggett bound. An upper bound on f_s computed from the density profile alone; for a weakly interacting condensate in a one-dimensional modulation it is an equality. In two dimensions upper and lower brackets exist.

Phase twist. Imposing a phase gradient on the order parameter, by twisted boundary conditions or a moving frame; the energy cost at small twist is proportional to f_s. Called the helicity modulus in lattice models.

Winding number. In path-integral quantum Monte Carlo, the number of times world lines wrap around the periodic box; its variance gives f_s.

Bogoliubov spectrum. The excitation spectrum of a weakly interacting condensate, $\varepsilon(k) = \sqrt{\epsilon_k(\epsilon_k + 2gn)}$ with $\epsilon_k = \hbar^2k^2/2m$: phonons with speed $c_s = \sqrt{gn/m}$ at small k, free particles at large k.

Healing length. $\xi = \hbar/\sqrt{2mgn}$ (conventions vary by $\sqrt{2}$): the length over which a condensate recovers from a local perturbation; the crossover scale of the Bogoliubov spectrum.

Quasi-condensate. A gas whose density fluctuations are suppressed while its phase still fluctuates over large distances; it looks bimodal in an image but need not be superfluid.

BKT transition. Berezinskii–Kosterlitz–Thouless transition: the superfluid transition of a two-dimensional gas, driven by the unbinding of vortex pairs; below it coherence decays algebraically and the superfluid density jumps to $n_s\lambda^2 = 4$ at T_c.

Phase-space density. $D = n\lambda^2$ with λ the thermal de Broglie wavelength; the BKT point is at $D_c = \ln(C/\tilde{g})$.

Two-dimensional coupling. The dimensionless constant $\tilde{g} = \sqrt{8\pi}\, a/\ell_z$, with a the scattering length and ℓ_z the oscillator length of the tight axis.

Quasi-2D. The regime where the tight-axis energy $\hbar\omega_z$ exceeds both k_BT and the interaction energy, so that the motion along z is frozen while collisions remain three-dimensional.

Accordion lattice. A standing wave made by two beams crossing at a variable angle; changing the angle changes the spacing and compresses a single plane of atoms.

Box trap. A potential that is flat inside and steep at the walls, made with shaped light (an intensity mask or a digital micromirror device, DMD); it gives a uniform gas.

Time of flight (TOF). Imaging after the trap is switched off and the cloud has expanded; in the far field the image reflects the in-trap momentum distribution.

Far field. The expansion time after which the initial size of the cloud no longer matters; for a coherent cloud it can be much longer than the flight times used.

Band mapping. Switching a lattice off slowly compared with the band gap but fast compared with tunnelling, so that each quasimomentum maps onto a free momentum and each band onto its Brillouin zone.

Bragg spectroscopy. Two laser beams with momentum difference $\hbar\mathbf{q}$ and frequency difference ω transfer atoms; the transfer measures the dynamic structure factor $S(\mathbf{q}, \omega)$.

Lattice-modulation spectroscopy. Periodic modulation of the lattice depth; the energy absorbed at each frequency reveals gaps and excitation bands.

Amplitude (Higgs) mode. The oscillation of the magnitude of the order parameter near the superfluid–Mott-insulator transition, whose gap softens at the critical point.

Bose–Hubbard model. Lattice model of bosons with nearest-neighbour hopping J and on-site repulsion U; valid for a deep lattice with a well-separated lowest band.

Mott insulator. The Bose–Hubbard phase at large U/J and integer filling: fixed site occupations, a gap and no long-range phase coherence.

Bose glass. An insulating but compressible, gapless phase of bosons in disorder, without long-range phase coherence.

Anderson localisation. Exponential localisation of non-interacting waves by interference of multiple scattering in a disordered potential.

Mobility edge. The energy that separates localised from extended single-particle states in a disordered potential (in three dimensions).

Speckle. The random intensity pattern of laser light passed through a diffuser; a disorder potential with a finite correlation length.

Bichromatic lattice. Two superposed lattices with incommensurate periods; a quasi-periodic potential that realises the Aubry–André model in one dimension.

Optical quasicrystal. A potential made of several lattices whose angles give long-range order but no periodicity, such as the eightfold lattice of four one-dimensional lattices at 45°.

Quantum-gas microscope. High-resolution fluorescence imaging of a lattice gas with single-site resolution; it detects the occupation of each site modulo two.

Josephson array. A chain or lattice of condensates coupled by tunnelling; the frequency of small oscillations gives the Josephson coupling, and in a supersolid the superfluid fraction.